Quantum Gravity Dynamics

Chapter 8 — Cosmology

Natural-invariant-first FLRW dynamics and QGD quantum corrections

Reconstruction of the historical `Ch5-cosmology.tex` against the established Chapters 1–6

Action-first / dark energy tests v2.2 | 5 November 2026

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Abstract

The historical QGD cosmology manuscript proposed that a homogeneous temporal natural invariant obeys an autonomous fourth-order wave equation whose growing mode generates inflation, whose kinetic sector produces an exact dark-energy equation of state $w = -1$, and whose attractor predicts the observed vacuum density. Those assertions do not follow from the corrected six-chapter QGD programme. Here we preserve their genuine research questions but reconstruct cosmology using the complete coframe/signed-sector/derivative master metric and the presently controlled metric-only effective action. We give an exact fixed-flat-reference FLRW lift, distinguish a background representation from source-selected Friedmann evolution, and recover the massless-Dirac radiation consistency benchmark. We prove that background factorization alone cannot determine $H(t)$ or $w(z)$; extend Chapter 6's energy-first analysis to exact FLRW Misner–Sharp energy, pressure work, radiation redshift, and the cosmological apparent horizon; prove that Chapter 5's static collective Poisson model cannot be used unchanged as a homogeneous matter-filled cosmology; audit the old scalar fourth-order equation and its purported modes; correct the kinetic equation-of-state claim and the numerical vacuum-density argument; and formulate distinct conditional mechanisms for dark energy, inflation, and cosmological quantum response. The QGD-specific programme is an independently computed quantum-state-to-natural-invariant map with physical source vertices and a healthy cosmological collective response, not the insertion of a preferred spin-two field, a universal σ_t potential, or an assumed fourth-order dynamics. The v1.2 results include an exact collisionless Dirac-gas homogeneous energy response and a zero-mode/causal-kernel boundary protocol. The v1.3 result extends the Chapter 1 scalar-contact model to a finite-temperature regulated Hartree branch, derives a conditional isentropic Friedmann history from its thermodynamics, and calculates its homogeneous collisionless retarded scalar inverse response. An exact positive subthreshold bound shows that the benchmark scalar channel contains no isolated real two-fermion-subthreshold pole. These are quantum-matter calculations and conditional effective-GR cosmology, not a QGD-specific microscopic derivation of Einstein gravity or accelerated expansion. The v1.4 result establishes the vacuum-shift ambiguity, derives the bare local Dirac heat-kernel curvature response in a separately rematched covariant scheme, calculates its adiabatic condensate susceptibility, and proves that a pure $R + \alpha R^2$ local truncation cannot select nonzero exact de Sitter curvature when the effective cosmological constant vanishes. The v1.6 analysis computes the leading shear-stress pair-production spectral function of the Chapter 1 Hartree Dirac quasiparticles, proves homogeneous-energy Ward suppression and exact scalar–tensor decoupling in an isotropic state, and finds no isolated graviton-like tensor pole in this controlled one-loop channel. The complete finite-momentum interacting stress response remains open. Version 1.9 makes signed natural-invariant cosmological variables primary, incorporates QGD quantum corrections from the renormalized action, and proves a normalized radiation-era first-order evolution law. The final v2.0 consolidation adds reference-epoch normalization across cosmic eras, a no-extra-equation theorem for the complete metric-only action, a singular-map counterexample, and an explicit criterion for genuinely independent QGD predictions. No measured cosmological constant or microscopic origin of Einstein gravity is predicted. The v2.2 addendum integrates Q055–Q057 sector-selection and growth tests and introduces Q058: a shift-symmetric generalized-metric condensate has an exact early-radiation asymptote that strongly constrains evolving late-time dark energy, whereas a bounded canonical hilltop requires an explicit physical sector reference. Neither is added to the canonical action.

Status contract. [D] = exact algebra, standard geometric identity, or explicit check under stated hypotheses; [C] = consequence of an adopted effective action, chosen state, or correspondence

assumptions; [A] = separately stated model hypothesis; [O] = not yet established for microscopic QGD. A verified representation of an existing Einstein solution is not an independent prediction of that solution.

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1 Historical source, physical aim, and a strict proof boundary

The earlier manuscript entitled *Chapter 5: Cosmology* is a historical idea source, not this research programme's governing derivation. Its appealing ambitions were to explain homogeneous expansion using collective matter-wave degrees of freedom, investigate an early accelerated phase and a late accelerated phase, and compare their observables with astronomical data. Those aims survive. The proposed autonomous equation

$$\Box_g \sigma_\mu = Q_\mu + G_\mu + T_\mu + \kappa \ell_Q^2 \Box_g^2 \sigma_\mu \tag{1}$$

is *not* the action-derived Euler equation of Chapters 2 and 4. The historical Q_μ current is not the symmetric master-metric operator $\mathcal{Q}_{ab}$, and the revised three-channel balance is an Euler derivative, not a source decomposition of $\Box_g \sigma_\mu$. The fourth-order operator, its sign, coefficient, covariance, and boundary data remain [O]. The older assertion that an instability saturates to drive inflation also remains [O].

The physical ambition is unchanged: quantum matter is fundamental in the proposed ontology, while Einsteinian geometry is a successful large-scale description. Chapter 1 supplies microscopic and macroscopic Dirac information; Chapter 2 supplies the metric representations and regularity tests; Chapter 3 calibrates Newtonian gravity; Chapter 4 gives the effective Euler equation and quantum stress; Chapter 5 establishes a *conditional static* matter-wave-to-gravity bridge; and Chapter 6 rules out interpreting redundant signed-sector amplitudes as independently physical vacuum energy. None of these chapters has yet shown that the microscopic matter action selects a unique cosmological collective mode, dark-energy history, or inflationary potential.

2 The governing QGD structures in cosmology

The *complete founding master metric* remains

$$\boxed{g_{\mu\nu} = T^a{}_\mu T^b{}_\nu \left[\eta_{ab} - \sum_{s=1}^{N} \varepsilon_s \sigma_a^{(s)} \sigma_b^{(s)} - \lambda \mathcal{Q}_{ab} \right], \qquad \varepsilon_s = \pm 1.} \tag{2}$$

Here $T^a{}_\mu$ is the reference coframe and $\sigma_a^{(s)}$ are promoted natural-invariant covectors, not universally equal to the free Dirac momentum ratios. The off-shell functional $\mathcal{Q}_{ab}[\sigma; T]$ and λ are not derived by the microscopic action. The commonly used fixed-coframe, $\lambda \mathcal{Q} = 0$ chart is a *specialization*, not a replacement for Eq. (2).

For a renormalized metric-only effective action $\Gamma_{\rm ren}[g, \text{matter}]$, define the measure-weighted metric Euler tensor Π^{ab} pulled to the reference frame. Its natural-invariant equations have the conditional complete-map form

$$\boxed{2 s_s \Pi^{ab} \sigma_b^{(s)} = \lambda \big[(D_s \mathcal{Q})^*_{\sqrt{-g}} \Pi \big]^a, \qquad s_s = -\varepsilon_s.} \tag{3}$$

The geometric pullback version selected in Chapter 6 is

$$\boxed{(D_{\sigma^{(s)}} g)^*_{\sqrt{-g}} \big(\Pi_{\mathcal{G}} + \Pi_{\rm cl} + \Pi_{\rm q} \big) = 0.} \tag{4}$$

When the metric-map differential spans all admissible metric variations and boundary terms are controlled, these equations impose the adopted effective Einstein–matter metric equations. They do not in themselves derive that adopted action from microscopic quantum matter. On a degenerate one-sector branch the converse can fail.

The quantum-corrected *solution* is denoted

$$\boxed{g_{\mu\nu}^{\rm ren} = g_{\mu\nu}^{(0)} + \delta g_{\mu\nu}^{\rm quantum} + O(\hbar^2).} \tag{5}$$

The response $\delta g^{\mathrm{quantum}}$ results from quantum matter stress in the renormalized effective equations. It is not a second independently varied $\mathcal{Q}_{ab}$ term. One may sometimes represent an on-shell correction through $-\lambda\mathcal{Q}^{\mathrm{rep}} = \delta g^{\mathrm{quantum}}$ by convention; the signed sectors can alternatively be readjusted with $\mathcal{Q} = 0$. Neither representation determines a unique off-shell $\mathcal{Q}$. This separation is mandatory in the cosmology chapter as well.

3 What homogeneity and isotropy really require

Use signature $(-,+,+,+)$ and initially set $c = \hbar = 1$. A spatially flat FLRW metric in cosmic time is

$$\boxed{\mathrm{d}s^2 = -\mathrm{d}t^2 + a^2(t)\delta_{ij}\mathrm{d}x^i\mathrm{d}x^j, \qquad a(t) > 0, \quad H = \dot{a}/a.} \tag{6}$$

An individual homogeneous four-vector respecting strict rotational invariance has vanishing spatial components in its comoving frame. It does *not* follow that every QGD signed-sector covector has only one common temporal component. An organized spatial triad or an eight-sector collection can produce an isotropic *composite metric* while its individual representatives transform among themselves. Moreover, with a fixed Cartesian flat coframe, one purely temporal quadratic one-form changes g_{tt} but leaves $g_{ij} = \delta_{ij}$; it cannot create the spatial scale factor $a(t)$.

3.1 Exact fixed-flat-reference FLRW lift

Proposition 3.1 (Constructive FLRW lift [D])**.** *Choose a fixed Cartesian flat reference* $\bar{g} = -\mathrm{d}t^2 + \sum_i(\mathrm{d}x^i)^2$ *and the controlled specialization* $\lambda\mathcal{Q} = 0$*. For any smooth* $a(t) > 0$ *put* $D(t) = a^2(t) - 1$ *and choose a nonzero constant* b*. The six real signed covectors*

$$\boxed{\sigma^{(i,\pm)} = \frac{D(t) \pm b^2}{2b}\,\mathrm{d}x^i, \quad i = 1,2,3} \tag{7}$$

satisfy

$$\sum_{i=1}^{3}\left(\sigma^{(i,+)} \otimes \sigma^{(i,+)} - \sigma^{(i,-)} \otimes \sigma^{(i,-)}\right) = (a^2 - 1)\sum_i \mathrm{d}x^i \otimes \mathrm{d}x^i. \tag{8}$$

Hence $\bar{g} + \sum_i(\sigma^{(i,+)}\sigma^{(i,+)} - \sigma^{(i,-)}\sigma^{(i,-)})$ *equals Eq. (6) exactly for every* $a(t) > 0$*.*

Proof. The elementary identity $[(D + b^2)^2 - (D - b^2)^2]/(4b^2) = D$ holds for all real D and $b \neq 0$. Apply it separately to each spatial diagonal component. No curvature is preassigned through the reference tetrad. □

Remark 3.2 (The lapse direction and regularity). Chapter 4's rank audit finds that the six-spatial-sector construction alone reaches nine independent metric perturbation directions; the lapse variation is absent. Adding an independent cancelling time pair $\sigma^{(0,+)} = v\mathrm{d}t$ and $\sigma^{(0,-)} = v\mathrm{d}t$ with $v \neq 0$ leaves the metric unchanged but permits a rank-ten regular lift. Equivalently, the exact all-four-pair conformal-time construction of Chapter 2 is spanning. A regular lift is needed to infer all metric Euler equations from sector variations; a background-only six-sector representation is insufficient for unrestricted perturbation dynamics.

3.2 Conformal-time lift with all four directions

In conformal time η, $g_{\mu\nu} = a^2(\eta)\eta_{\mu\nu}$. Define $d_a = (a^2 - 1)\eta_{aa}$ (no sum) and $\omega \neq 0$. Then

$$\boxed{\sigma^{(a,\pm)} = \frac{d_a \pm \omega^2}{2\omega}\,\mathrm{d}x^a, \quad \eta_{\mu\nu} + \sum_{a=0}^{3}\left[\sigma_\mu^{(a,+)}\sigma_\nu^{(a,+)} - \sigma_\mu^{(a,-)}\sigma_\nu^{(a,-)}\right] = a^2\eta_{\mu\nu}.} \tag{9}$$

This is the existing Chapter 2 QGD–067 consistency construction. The arbitrary b and ω are *factorization parameters*, not new Hubble-scale couplings, condensate order parameters, or fitted quantum gravitational constants.

Theorem 3.3 (No expansion law from metric factorization alone [D])**.** *On the fixed-reference $\lambda\mathcal{Q} = 0$ branch, the algebraic existence of signed-sector FLRW covectors does not constrain the function $a(t)$ or determine $H(t)$, acceleration, a cosmological equation of state, or an inflationary history.*

Proof. Proposition 3.1 constructs a real signed-sector configuration for *every* smooth positive $a(t)$, including dust, radiation, exponential acceleration, deceleration, and arbitrary smooth interpolations. Therefore the algebraic representability condition alone cannot select any one of those expansion histories. The action, physical source, state, and boundary/initial conditions must supply the selection. □

4 Effective cosmological dynamics and conservation

The presently controlled metric-only action yields the usual Einsteinian equations when the spanning and matter-on-shell assumptions hold. Let ϵ denote the *physical total energy density* and p the isotropic pressure, both with SI units J m^{-3} when c is restored. Keep any cosmological constant Λ as an *effective-action parameter*, not a QGD microscopic prediction. For spatially flat FLRW,

$$\boxed{H^2 = \frac{8\pi G_N}{3c^2}\epsilon + \frac{\Lambda c^2}{3}, \qquad \dot{H} = -\frac{4\pi G_N}{c^2}(\epsilon + p).} \tag{10}$$

The acceleration and continuity equations are

$$\boxed{\frac{\ddot{a}}{a} = -\frac{4\pi G_N}{3c^2}(\epsilon + 3p) + \frac{\Lambda c^2}{3}, \qquad \dot{\epsilon} + 3H(\epsilon + p) = 0.} \tag{11}$$

The conservation identity is the homogeneous form of the covariant matter Ward identity and contracted Bianchi identity; it is not a new separate σ_t energy-exchange equation. If individual fluid components exchange energy, $\dot{\epsilon}_A + 3H(\epsilon_A + p_A) = J_A$ with $\sum_A J_A = 0$; a specific nonzero J_A requires a microscopic interaction or matching calculation.

For constant equations of state $w_A = p_A/\epsilon_A$ of separately conserved species,

$$\epsilon_A(a) = \epsilon_{A,0}\, a^{-3(1+w_A)}, \qquad w_{\text{dust}} = 0, \quad w_{\text{radiation}} = \tfrac{1}{3}, \quad w_\Lambda = -1. \tag{12}$$

The familiar $a \propto t^{2/3}$ (dust), $a \propto t^{1/2}$ (radiation), and $a \propto e^{Ht}$ (positive vacuum-energy dominated) scale factors are [C] solutions of the adopted Einstein–source dynamics. They are exactly representable by Eq. (7) but are not predictions from the signed decomposition alone.

4.1 The observed dark-energy scale is not predicted by substituting H_0

A common historical mistake was to set $\epsilon_\sigma = \epsilon_{\text{crit}}$ by inserting an observed H_0 into the Friedmann equation and announce a prediction of vacuum energy. Instead, define

$$\boxed{\epsilon_{\text{crit},0} = \frac{3c^2 H_0^2}{8\pi G_N}, \qquad \epsilon_{\text{DE},0} = \Omega_{\text{DE},0}\, \epsilon_{\text{crit},0}.} \tag{13}$$

These are definitions within a given Friedmann model. For an illustrative $H_0 \simeq 2.2 \times 10^{-18}$ s^{-1}, $\epsilon_{\text{crit},0} \simeq 7.8 \times 10^{-10}$ J m^{-3}, and a fitted $\Omega_{\text{DE},0} \simeq 0.69$ gives $\epsilon_{\text{DE},0} \simeq 5.4 \times 10^{-10}$ J m^{-3}. The historical $3H_0^2c^4/(8\pi G_N)$ is dimensionally wrong for an energy density when H_0 is in inverse seconds; the correct numerator contains c^2. No equation in Chapters 1–6 computes $\Omega_{\text{DE},0}$ from microscopic QGD.

5 A physical quantum-matter cosmology already available

The strongest existing microscopic-to-cosmological *consistency benchmark* is the massless Dirac radiation construction in Chapter 2, not a single σ_t kinetic term. For $g_{\mu\nu} = a^2(\eta)\eta_{\mu\nu}$, including the curved-spacetime Dirac spin connection, the conformal field rescaling gives

$$\boxed{\Psi(\eta, \boldsymbol{x}) = a^{-3/2}(\eta)\chi(\eta, \boldsymbol{x}), \qquad i\gamma^\mu \partial_\mu \chi = 0,} \tag{14}$$

for the free massless branch. A homogeneous isotropic thermal ensemble of one full massless Dirac species has, in natural units,

$$\boxed{\epsilon_D(a) = \frac{\epsilon_{D,0}}{a^4}, \qquad p_D = \frac{1}{3}\epsilon_D, \qquad \epsilon_{D,0} = \frac{7\pi^2}{60}T_0^4.} \tag{15}$$

With $\Lambda = 0$ and this radiation source, the adopted Friedmann equation implies in conformal time

$$\boxed{a(\eta) = \sqrt{\frac{8\pi G_N \epsilon_{D,0}}{3}}\,(\eta - \eta_0), \qquad \eta > \eta_0,} \tag{16}$$

where units are $c = \hbar = k_B = 1$ and the normalization of a is fixed consistently. Equation (9) then provides the corresponding exact QGD signed-sector metric representation.

What is derived, and what is not. The conformal Dirac reduction and signed-factorization algebra are explicit. The thermal occupation and Einstein–Friedmann evolution are separately specified premises. A nonzero trace anomaly, state-dependent vacuum polarization, massive finite-density fermions and interactions can alter the effective $\langle T_{\mu\nu}\rangle_{\rm ren}$ and require further computation. Neither the Dirac gas nor the regular signed lift yet selects a collective gravitational mode from the *gravity-free* microscopic theory.

6 Quantum corrections, vacuum energy, and no double counting

The renormalized quantum matter contribution enters the effective cosmological Einstein equations through the consistently conserved stress tensor $T_{\rm q}^{\mu\nu}$, subject to the matter state and renormalization scheme. Its first-order metric response is schematically

$$\boxed{\mathcal{L}_{g^{(0)}}[\delta g^{\rm quantum}]_{\mu\nu} = \frac{8\pi G_N}{c^4}\langle \widehat{T}_{\mu\nu}\rangle^{(1)}_{\rm ren},} \tag{17}$$

with appropriate gauge and causal/boundary conditions. This is Chapter 5's controlled *solution-level* backreaction. The historical independent $-\lambda\mathcal{Q}_{ab}$ term does not need to be activated again to represent this same correction.

In an exact Lorentz-invariant vacuum, a renormalized stress proportional to the metric has $T^{\rm vac}_{\mu\nu} = -\epsilon_{\rm vac} g_{\mu\nu}$ in natural units (subject to sign conventions and cosmological-constant counterterms) and hence $p_{\rm vac} = -\epsilon_{\rm vac}$. But no Chapter 1–6 computation fixes the required positive residual cosmic vacuum energy; the regulated Chapter 6 Hartree grand potential has scheme-dependent vacuum subtraction and does not predict $\Lambda_{\rm obs}$. Its tested scalar Hessian is positive and nonzero, rather than containing the long-range static pole needed for gravity.

Proposition 6.1 (Cosmological sector-energy nonidentifiability [D] under metric-only dynamics)**.** *In the regular signed-sector metric-only theory, the background functions $\sigma_a^{(s)}(t)$ and their coordinate derivatives are not by themselves observable cosmological energy densities. In particular, the exactly cancelling time-dependent representatives $P(t) = N(t) = ve^{\theta(t)}I_4$ leave the metric unchanged while generally changing $\dot{P}$ and $\dot{N}$.*

Proof. For any smooth $\theta(t)$, $P^{\mathsf{T}}P - N^{\mathsf{T}}N = 0$. Its first time derivatives are $\dot{P} = \dot{N} = ve^{\theta}\dot{\theta}I_4$. Thus any positive naive quantity $\|\dot{P}\|_F^2 + \|\dot{N}\|_F^2 = 8v^2e^{2\theta}\dot{\theta}^2$ is representation-dependent although all metric observables are unchanged. Chapter 6 identifies the corresponding local metric-preserving $O(4,4)$ orbit. □

This result directly invalidates the historical identification of an arbitrary $\frac{1}{2}\dot{\sigma}_t^2$ with a uniquely physical cosmological dark-energy density. A separate microscopic sector-sensitive action could change the result, but its observable content, stability and constraints must then be established independently.

7 Chapter 6 energy equations in an expanding universe

Chapter 6 distinguishes microscopic flat-space Noether energy, renormalized matter stress, and geometric or boundary charges of the adopted effective metric. Cosmology cannot reuse the static Schwarzschild Brown–York expression with r replaced by $a(t)$: a generic FLRW background has no global timelike Killing symmetry, and a Brown–York energy depends on the observer, boundary geometry, and reference subtraction. Spherical symmetry nevertheless supplies a precise *Misner–Sharp* quasilocal energy [5, 6]; this is the controlled bridge for Chapter 7.

7.1 Exact FLRW quasilocal energy and pressure work

Proposition 7.1 (FLRW Misner–Sharp and comoving energy-work identities [D]|[C])**.** *Assume a spatially flat FLRW line element with cosmic time t, a comoving sphere of coordinate radius r_0, and physical areal radius $R(t) = a(t)r_0$. The geometric Misner–Sharp energy is*

$$\boxed{E_{\mathrm{MS}}(t,R) = \frac{c^4 R}{2G_N}\left(1 - g^{AB}\partial_A R\,\partial_B R\right) = \frac{c^2H^2R^3}{2G_N}.} \tag{18}$$

If, in addition, the adopted Einstein–Friedmann equation holds, with any cosmological constant absorbed into the total effective source $\epsilon_{\mathrm{tot}} := \epsilon + \epsilon_\Lambda$, $p_{\mathrm{tot}} := p - \epsilon_\Lambda$ and $\epsilon_\Lambda := \Lambda c^4/(8\pi G_N)$, then

$$\boxed{E_{\mathrm{MS}} = \frac{4\pi R^3}{3}\epsilon_{\mathrm{tot}} = V\epsilon_{\mathrm{tot}}, \qquad \dot{E}_{\mathrm{MS}}\big|_{r_0} = -p_{\mathrm{tot}}\dot{V}, \qquad V = \frac{4\pi R^3}{3}.} \tag{19}$$

Here the time derivative is taken along a comoving *spherical boundary. The first equality in Eq. (18) is geometric; the identification with $V\epsilon_{\mathrm{tot}}$ and the work law depend on the adopted field equations and conservation.*

Proof. On the (t,r) orbit space $\mathrm{d}s^2_{(2)} = -c^2\mathrm{d}t^2 + a^2\mathrm{d}r^2$, $\partial_t R = HR$ and $\partial_r R = a$. Consequently $g^{AB}\partial_A R\partial_B R = 1 - H^2R^2/c^2$ and Eq. (18) follows. The first Friedmann equation gives $3c^2H^2/(8\pi G_N) = \epsilon_{\mathrm{tot}}$, hence $E_{\mathrm{MS}} = V\epsilon_{\mathrm{tot}}$. Finally $\dot{V} = 3HV$ and $\dot{\epsilon}_{\mathrm{tot}} = -3H(\epsilon_{\mathrm{tot}} + p_{\mathrm{tot}})$ imply $\dot{E}_{\mathrm{MS}} = V\dot{\epsilon}_{\mathrm{tot}} + \epsilon_{\mathrm{tot}}\dot{V} = -p_{\mathrm{tot}}\dot{V}$. □

A precise radiation/energy benchmark. For a comoving sphere, a separately conserved component with constant $w_A = p_A/\epsilon_A$ has

$$\boxed{E_A(a) = V_0\epsilon_{A,0}\,a^{-3w_A}, \qquad \frac{\mathrm{d}E_A}{\mathrm{d}t} = -3Hw_AE_A, \qquad V_0 = \frac{4\pi r_0^3}{3}.} \tag{20}$$

Therefore dust ($w = 0$) has constant enclosed energy; Chapter 2's free massless Dirac radiation ($w = 1/3$) has $E_D \propto a^{-1}$; and a constant vacuum source ($w = -1$) has $E_{\mathrm{vac}} \propto a^3$. These statements are relative to a comoving volume and its stress-energy work balance. None assigns an independently

conserved energy to each signed-sector amplitude or predicts the vacuum energy magnitude. In the radiation case, combining Eq. (15) with Eq. (20) checks *both* the microscopic conformal scaling and the cosmological pressure-work identity.

7.2 Comoving energy versus apparent-horizon energy

For an expanding ($H > 0$), spatially flat FLRW spacetime the cosmological apparent-horizon areal radius is $R_A = c/H$. The same geometric charge at that *moving* boundary is

$$\boxed{E_A = \frac{c^4 R_A}{2G_N} = \frac{c^5}{2G_N H}, \qquad \dot{E}_A = -\frac{c^5}{2G_N}\frac{\dot{H}}{H^2}.} \tag{21}$$

For exact de Sitter expansion $H = \text{const}$, E_A is constant, whereas a comoving sphere encloses a growing vacuum energy $E_{\text{vac}} \propto a^3$. There is no contradiction: these are *different moving boundaries.* Equation (19)'s simple work law applies to the comoving sphere; it must not be substituted unchanged for flux through the apparent horizon. The covariant unified first law or a carefully referenced Brown–York prescription is needed for other boundary motions [5, 6, 7].

7.3 An energy reconstruction theorem—and its limit

Proposition 7.2 (Arbitrary FLRW histories admit a conserved effective source [D]|[C])**.** *For any sufficiently smooth positive scale factor $a(t)$, the effective energy density and pressure reconstructed by the* adopted spatially flat Einstein–Friedmann equations*, with Λ absorbed, are*

$$\boxed{\epsilon_{\text{rec}}(t) = \frac{3c^2H^2}{8\pi G_N}, \qquad p_{\text{rec}}(t) = -\frac{c^2}{8\pi G_N}\left(2\dot{H} + 3H^2\right).} \tag{22}$$

These obey $\dot{\epsilon}_{\text{rec}} + 3H(\epsilon_{\text{rec}} + p_{\text{rec}}) = 0$ identically. Wherever $H \neq 0$, their effective equation-of-state parameter is

$$\boxed{w_{\text{rec}} = -1 - \frac{2\dot{H}}{3H^2}.} \tag{23}$$

Thus a geometric energy dictionary plus an exact signed-sector FLRW lift does not by itself determine a physical cosmological source or choose a unique $a(t)$.

Proof. Insert Eq. (22) into $\dot{\epsilon} + 3H(\epsilon + p)$, noting $\dot{\epsilon} = 3c^2H\dot{H}/(4\pi G_N)$ and $\epsilon + p = -c^2\dot{H}/(4\pi G_N)$. The two terms cancel. Since Proposition 3.1 constructs a metric representative for every such $a(t)$, requiring a compatible formal effective source cannot select a history without a microscopic equation of state, an independently defined collective mode, initial conditions and physical constraints. The reconstructed source need not correspond to any healthy microscopic quantum state. □

Physical filters, not predictions. In flat Einstein cosmology, $\epsilon_{\text{rec}} + p_{\text{rec}} \geq 0$ is equivalent to $\dot{H} \leq 0$ (null energy condition). Acceleration, $\ddot{a} > 0$, requires $\epsilon_{\text{rec}} + 3p_{\text{rec}} < 0$ for positive total density. These tests can exclude proposed source actions, but do not uniquely select an inflationary phase or late-time dark energy. If one extracts w_{rec} from measured $H(z)$, it remains a reconstruction, not a zero-parameter prediction of QGD.

8 The missing cosmological zero mode of Chapter 5's static collective bridge

This is the direct cross-chapter test of Chapter 5's conditional matter-wave energy functional, distinct from a postulated cosmological scalar action.

Proposition 8.1 (Homogeneous-source obstruction for the unmodified static bridge [D]|[C])**.** *Consider the conditional Chapter 5 functional on a spatial Euclidean slice,*

$$\boxed{E_{\mathrm{stat}}[q;\rho_m] = \int \mathrm{d}^3x_{\mathrm{phys}} \left[\frac{Z}{2}|\nabla_{\mathrm{phys}} q|^2 - g_c q \rho_m\right], \qquad Z > 0, \quad g_c \neq 0.} \tag{24}$$

Its static Euler equation is $-Z\nabla^2_{\mathrm{phys}} q = g_c \rho_m$. It has no spatially homogeneous solution $q = q(t)$ for a nonzero spatially homogeneous source $\rho_m = \bar{\rho}_m(t) \neq 0$ at a given time. Moreover, on a compact flat spatial slice with periodic boundary conditions, no smooth periodic static solution exists if $\int \rho_m \mathrm{d}^3x_{\mathrm{phys}} \neq 0$.

Proof. For a spatially homogeneous q, $\nabla^2 q = 0$, so the static Euler equation requires $g_c \bar{\rho}_m = 0$, contrary to the hypotheses. For a periodic compact slice, integration of the Laplacian gives zero while integration of the homogeneous/nonzero-mean source does not. In noncompact Euclidean space, $q_{\mathrm{part}} = -(g_c \bar{\rho}_m/(6Z))R^2$ formally solves the equation for constant $\bar{\rho}_m$, but is spatially nonhomogeneous and unbounded; it does not supply a homogeneous finite-energy cosmological order parameter. □

Interpretation and boundary. This is a no-go result for using the *unchanged, static, scalar Chapter 5 ansatz* as the entire homogeneous cosmological field equation. It is *not* a no-go theorem for QGD, FLRW expansion, nonlocal response, temporal collective dynamics or cosmological perturbations. A familiar conditional perturbative separation is $\rho_m = \bar{\rho}_m + \delta\rho_m$ and

$$\boxed{\nabla^2_{\mathrm{com}}\, \delta q = -\frac{g_c a^2}{Z}\, \delta\rho_m, \qquad \int_{\mathrm{compact}} \delta\rho_m \, \mathrm{d}^3x = 0,} \tag{25}$$

provided one explicitly *postulates* the background subtraction and interprets δq as a fluctuation, rather than claiming the subtraction follows from Chapter 5's original static action. On a noncompact slice, boundary/data restrictions replace the periodic integral constraint. Equation (25) can address inhomogeneous static response in a specified cosmological setting; it *does not* determine the background $H(t)$.

To extend Chapter 5 to the cosmic background one must derive or independently specify the temporal response kernel, its quantum-state dependence, correct stress energy and Ward identities, boundary/zero-mode conditions, and the physical projection from collective observables to natural-invariant equivalence classes. Simply adding $Z_t \dot{q}^2/2$, a potential, or a friction term to the Chapter 5 functional would be a [A] trial unless matched to a microscopic calculation. In particular, Chapter 6's $O(4,4)$ metric-null directions cannot be assigned physical positive kinetic energy by this expedient.

9 Audit of the historical fourth-order cosmology

Equation (1) is not the QGD Euler equation and contains no specified physical kinetic action from which a homogeneous cosmological mode spectrum can be deduced. To show how stringent a *consistent hypothetical* higher-derivative test would be, consider a scalar $\varphi(t)$ on a prescribed FLRW background. Define $\mathscr{D} = \partial_t^2 + 3H\partial_t$ and signature $(-,+,+,+)$ so that $\Box_g \varphi = -\mathscr{D}\varphi$ for a homogeneous scalar. Direct calculation gives

$$\boxed{\begin{aligned}\Box_g^2 \varphi &= \mathscr{D}^2 \varphi \\ &= \varphi^{(4)} + 6H\varphi^{(3)} + (9H^2 + 6\dot{H})\ddot{\varphi} + (9H\dot{H} + 3\ddot{H})\dot{\varphi}.\end{aligned}} \tag{26}$$

This is an exact scalar identity. It is *not* automatically the tensor/covector operator $\Box_g^2 \sigma_\mu$; for a covector, connections and curvature terms must also be included. The fourth-derivative bracket

written in the historical manuscript is not Eq. (26), nor does its claimed de Sitter factorization follow from its displayed equation.

For illustration only, *postulate* a different, mathematically explicit scalar equation

$$\boxed{\mathscr{D}(1-\ell^2\mathscr{D})\varphi = 0, \qquad \ell > 0,} \tag{27}$$

with ℓ a time scale (a length divided by c in SI units). If $H = H_0$ is fixed, the characteristic roots are

$$\boxed{r_0 = 0, \qquad r_1 = -3H_0, \qquad r_\pm = \frac{-3H_0 \pm \sqrt{9H_0^2 + 4/\ell^2}}{2}.} \tag{28}$$

One extra root is positive for $H_0 \geq 0$. It represents a linear growing solution in this *trial* equation, not a demonstrated healthy inflationary mode. The old $e^{\pm t/\ell}$ values omit the Hubble-friction modification; they are only leading limits when $H_0\ell \ll 1$. No interpretation as an Ostrogradsky ghost, a stabilized condensate, or an inflationary solution follows without a complete action, Hamiltonian and nonlinear constraint analysis. In particular the historical $\sigma_{\rm sat} \sim \ell_{\rm Pl}$ compares a putatively dimensionless field to a length without defining its dimensions.

10 Correct equations of state and viable candidate mechanisms

The historical $\rho = +\frac{1}{2}\dot{\sigma}_t^2$, $p = -\frac{1}{2}\dot{\sigma}_t^2$ is not the canonical stress tensor of a homogeneous scalar with only a standard positive kinetic term. To evaluate a possible *new physical collective scalar* φ distinctly from redundant σ_t, take the illustrative effective action (natural units)

$$S_\varphi = -\int \mathrm{d}^4x\sqrt{-g}\left[\frac{Z}{2}g^{\mu\nu}\partial_\mu\varphi\partial_\nu\varphi + V(\varphi)\right], \qquad Z > 0. \tag{29}$$

This is an [A] phenomenological toy action, *not* derived from the microscopic QGD model and *not* an independent new fundamental gravitational sector. Its homogeneous stress tensor obeys

$$\boxed{\epsilon_\varphi = K + V, \qquad p_\varphi = K - V, \qquad K = \frac{Z}{2}\dot{\varphi}^2 \geq 0, \qquad w_\varphi = \frac{K-V}{K+V}.} \tag{30}$$

For $V = 0$, $w = +1$ (stiff matter), *not* -1; if $K \ll V$ and $V > 0$, then $w \approx -1$; if $K = 0$ exactly, $w = -1$. For positive $K + V$, $w + 1 = 2K/(K+V) \geq 0$. A phantom kinetic sign would introduce a different stability and positivity problem rather than making a kinetic-only $w = -1$ law. A healthy accelerated scalar-dominated branch requires $V > 2K$ because $\epsilon_\varphi + 3p_\varphi = 4K - 2V < 0$.

The field equation in this specific toy is

$$\boxed{Z(\ddot{\varphi} + 3H\dot{\varphi}) + V'(\varphi) = 0,} \tag{31}$$

with additional interactions only if separately specified. It demonstrates what *must* be calculated to predict a cosmological $w(z)$: the correct kinetic operator, the normalized potential, an initial state and the gravitational background response.

The archival polynomial $V_{\rm old}(\sigma_t) = \sigma_t^2/8 - \sigma_t^4/16$ is retained solely as a proposed *shape* whose microscopic origin is missing. It carries no stated physical energy-density normalization, and σ_t is not invariant under the signed-sector redundancies of Chapter 6. Therefore it is not a derived universal QGD potential, cannot by itself give an absolute w_0, w_a, and should not be inserted into Eq. (31) without a physical collective variable and dimensional matching. The assertion that it is uniquely selected by the static Schwarzschild sector is also superseded by Chapter 2's source-conditioned metric Euler analysis.

10.1 An honest route to dynamical dark energy

An independently derived cosmological source or effective collective mode could have a time-dependent $w(a)$, in which case homogeneous conservation yields

$$\boxed{\epsilon_{\rm DE}(a) = \epsilon_{\rm DE,0} \exp\left[3\int_a^1 (1+w(\tilde{a}))\,\mathrm{d}\ln\tilde{a}\right].} \tag{32}$$

The corresponding flat-background relation is

$$\boxed{\frac{H^2(a)}{H_0^2} = \Omega_{r0}a^{-4} + \Omega_{m0}a^{-3} + \Omega_{\rm DE,0}\frac{\epsilon_{\rm DE}(a)}{\epsilon_{\rm DE,0}}, \qquad a = (1+z)^{-1}.} \tag{33}$$

The DESI DR2 cosmology results [4] find that some combinations of BAO, CMB and supernova data favor time-varying phenomenological dark energy relative to ΛCDM, with significance depending on data combination and modelling; the evidence does not validate the archival σ_t potential. The actual observational prediction must follow from a derived $w(a)$, and perturbations must be consistent with background fits. There is currently no zero-extra-parameter QGD curve to fit.

11 Inflation as a separate dynamical problem

Acceleration requires $\epsilon + 3p < \Lambda c^4/(4\pi G_N)$ in Eq. (11); with $\Lambda = 0$ this reduces to $w_{\rm total} < -1/3$ when $\epsilon > 0$. An inflationary phase must additionally solve the horizon and flatness requirements, supply sufficient accelerated expansion, end consistently, reheat into known matter, and generate perturbations compatible with cosmological data. Define

$$\boxed{\epsilon_H = -\frac{\dot{H}}{H^2}, \qquad N_e = \int_{t_i}^{t_f} H\,\mathrm{d}t.} \tag{34}$$

For $H > 0$, acceleration requires $\epsilon_H < 1$; sustained inflation requires a suitable history of this quantity, not merely an exponential solution for an auxiliary field on a preassigned de Sitter background. The archival claim that a fourth-order growing mode necessarily produces inflation and saturates after a Planck time is unproved. A complete candidate must provide an action and derive a gauge-invariant scalar/tensor perturbation spectrum. Standard slow-roll formulas for n_s and r cannot be applied directly to the unnormalized $V_{\rm old}(\sigma_t)$ or to a noncanonical, unidentified natural-invariant component.

12 Perturbations and tests that would distinguish QGD

A successful model must be compared against at least four observable classes: expansion distances, structure growth, CMB acoustic and lensing observables, and gravitational-wave propagation. For spatial flatness,

$$D_M(z) = c\int_0^z \frac{\mathrm{d}z'}{H(z')}, \qquad D_L(z) = (1+z)D_M(z). \tag{35}$$

The acoustic angular scale depends on $D_M(z_*)$ and the self-consistently calculated sound horizon $r_s(z_*)$, $\ell_A = \pi D_M(z_*)/r_s(z_*)$. Consequently the historical estimate $\Delta\ell_1/\ell_1 \approx -f_\sigma/2$ is not a generic derived QGD prediction; a full early-universe expansion and perturbation calculation is needed. Likewise a candidate $w(z)$ must generate physically normalized perturbations, source couplings to photons and matter, and a consistent gravitational slip and wave-speed sector rather than being fitted only at background level.

On a regular branch with a metric-only Einstein effective action, the metric and gravitational-wave sector at the same sources are observationally Einsteinian. A genuinely new QGD prediction must come from independently computed microscopic constraints on states, vertices or quantum corrections, or a new nonredundant sector action which survives consistency tests. Purely re-expressing the same FLRW solution in signed natural invariants does not change observable distances or generate a new acoustic peak shift.

13 A QGD-specific microscopic cosmology research programme

The constructive programme from Chapters 1, 5 and 6 specifies the missing dynamical arrows:

$$\boxed{\text{quantum state } \varrho \;\longrightarrow\; \varphi^I[\varrho] \;\longrightarrow\; [\Sigma] \;\longrightarrow\; g_{\mu\nu}[\Sigma, T, \mathcal{Q}] \;\longrightarrow\; \text{cosmological observables.}} \tag{36}$$

The collective observables φ^I must be genuine renormalized, appropriately coarse-grained operators of the chosen microscopic matter theory; $[\Sigma]$ means natural-invariant classes modulo redundancies. Given a specified homogeneous/thermal or out-of-equilibrium quantum state, compute its connected source response and the effective collective Hessian

$$\boxed{K_{IJ}(\omega, \boldsymbol{k}; t) = \frac{\delta^2 \Gamma_{\text{coll}}}{\delta\varphi^I(-k)\delta\varphi^J(k)}, \qquad \chi_{AB}(k) = V_A{}^I [K^{-1}]_{IJ} V_B{}^J + \chi_{AB}^{\text{contact}}.} \tag{37}$$

In a nonstationary FLRW setting the Fourier-frequency notation is an adiabatic/Wigner shorthand; the exact response is a two-time kernel with causal boundary conditions. The source vertices V_A must be calculated for rest energy, stress, radiation and independent probes, not assumed universal. A long-range static pole and healthy relativistic tensor sector are stringent *tests* of microscopic selection, not starting axioms. Perturbative constraints and the metric-map Jacobian must be evaluated on the chosen cosmological background.

We propose four sharply scoped next steps, without claiming they are completed:

1. **Microscopic equation of state and correlators.** Extend Chapter 1's controlled Dirac/Maxwell/contact model from a free radiation benchmark to a specified interacting finite-temperature/density state. Compute $\epsilon(T, \mu)$, $p(T, \mu)$, $\langle T_{\mu\nu} T_{\alpha\beta} \rangle_c$ and Ward identities with consistent renormalization; compute the homogeneous source response separately from the $\boldsymbol{k} \neq 0$ fluctuation susceptibility, in light of Proposition 8.1.
2. **Physical cosmological sector selection.** Find an operational map from these observables to $[\Sigma]$ that does not merely factorize a presolved Einstein metric, and test which signed-sector redundancies remain unobservable.
3. **Dynamics and stability.** Compute K_{IJ} and source vertices in the chosen expanding state. Look for real healthy propagating poles, positivity, correct gradient signs, causal propagation, and universal source response, checking the Chapter 5 static limit.
4. **Predictions, only after derivation.** Insert the resulting controlled quantum stress/collective dynamics into background and linear perturbation equations. Compute $H(z)$, $w(z)$, growth, D_M, CMB source functions, n_s, r, and gravitational-wave observables; compare with properly stated observational likelihoods.

14 The first homogeneous response calculation from Chapter 1 matter

The background problem can be sharpened without assuming that a single natural-invariant amplitude drives the universe. Here the microscopic input is the *free, positive-energy, collisionless*

Dirac kinetic limit of Chapter 1, propagated on a *given* FLRW geometry. The quantum mode-production, renormalized vacuum, contact-interaction, and electromagnetic contributions are not silently absorbed into this calculation. Set $c = \hbar = 1$ in this section, and let $q = a|\boldsymbol{p}_{\rm phys}|$ be conserved comoving momentum along freely falling collisionless trajectories. Choose a nonnegative isotropic occupation $f(q)$ (including relevant particles and antiparticles according to the degeneracy g_D), independent of time in this approximation, and an unchanging physical mass $m \geq 0$.

Proposition 14.1 (Exact kinetic homogeneous response [D]|[C])**.** *Assume smooth $a(t) > 0$, collisionless on-shell transport with $\partial_t f(q) = 0$, and finite moments. The particle/kinetic contributions to homogeneous energy density and isotropic pressure are exactly*

$$\boxed{\begin{aligned}\epsilon_D(a) &= \frac{g_D}{2\pi^2 a^4}\int_0^\infty q^2\sqrt{q^2+a^2m^2}\, f(q)\,\mathrm{d}q,\\ p_D(a) &= \frac{g_D}{6\pi^2 a^4}\int_0^\infty \frac{q^4}{\sqrt{q^2+a^2m^2}}\, f(q)\,\mathrm{d}q.\end{aligned}} \tag{38}$$

They obey the exact kinetic identities

$$\boxed{\begin{aligned}&\epsilon_D - 3p_D = mn_s \geq 0, \qquad n_s = \frac{g_D m}{2\pi^2 a^2}\int_0^\infty \frac{q^2 f(q)}{\sqrt{q^2+a^2m^2}}\,\mathrm{d}q,\\ &\frac{\mathrm{d}\epsilon_D}{\mathrm{d}\ln a} = -3(\epsilon_D + p_D), \qquad 0 \leq \frac{p_D}{\epsilon_D} \leq \frac{1}{3} \quad (\epsilon_D > 0).\end{aligned}} \tag{39}$$

The derivative is a homogeneous kinematic/adiabatic response at fixed comoving occupation*, not the full frequency-dependent retarded susceptibility of a quantum interacting state.*

Proof. The physical momentum measure gives $\mathrm{d}^3p_{\rm phys} = a^{-3}\mathrm{d}^3q$ and one-particle energy $E_q = a^{-1}\sqrt{q^2+a^2m^2}$, yielding Eq. (38). Subtracting three times the pressure gives m^2a^2 times the shared positive integrand and establishes $\epsilon_D - 3p_D = mn_s$. Differentiating $a^{-4}\sqrt{q^2+a^2m^2}$ under the integral gives $-4\sqrt{\cdots} + a^2m^2/\sqrt{\cdots} = -3\sqrt{\cdots} - q^2/\sqrt{\cdots}$, which is $-3(\epsilon_D + p_D)$. Finally $0 \leq q^2/(q^2+a^2m^2) \leq 1$ establishes $0 \leq 3p_D/\epsilon_D \leq 1$ for any nonnegative f. □

Corollary 14.2 (No acceleration from the isolated free kinetic Dirac source [C])**.** *If* in addition *the spatially flat Einstein–Friedmann equations of Section 4 are adopted, $\Lambda = 0$, and the* entire *physical source is the nonzero positive collisionless Dirac kinetic stress in Proposition 14.1, then*

$$\boxed{\dot{H} = -4\pi G_N(\epsilon_D + p_D) < 0, \qquad \frac{\ddot{a}}{a} = -\frac{4\pi G_N}{3}(\epsilon_D + 3p_D) < 0.} \tag{40}$$

Thus this controlled branch does not generate inflation or late-time acceleration. The result neither rules out other quantum states, a separate vacuum stress, interactions, a genuine collective mode, nor a non-Einstein effective action.

For $m = 0$, the integrals give $p_D = \epsilon_D/3$ and $\epsilon_D \propto a^{-4}$, reproducing the conformal massless-Dirac radiation benchmark already checked in Chapters 2 and 6. For am much larger than the occupied comoving momenta, $p_D/\epsilon_D \to 0$ and $\epsilon_D \sim a^{-3}mn_0$. With $f(q) = [e^{q/T_0}+1]^{-1}$ and $g_D = 4$ at $m = 0$, the standard integral gives $\epsilon_{D,0} = 7\pi^2T_0^4/60$. These are transparent matter benchmarks, not independent derivations of the effective geometry.

Essential scope restriction. For a genuinely massive *quantum* Dirac field on time-dependent FLRW, nonadiabatic particle production and vacuum polarization can invalidate the assumption $\partial_t f = 0$ in a given particle basis. Vacuum subtraction and trace anomalies may contribute to the *renormalized* source even for massless conformal matter. The kinetic result cannot be used to declare a universal "fermions never accelerate the universe" theorem. FLRW renormalization for quantum Dirac fields has an explicit adiabatic formalism [23, 24].

15 Zero-momentum boundary conditions: cosmological background versus perturbation

Proposition 8.1 shows exactly where the Chapter 5 static Newtonian model fails if it is used for a homogeneous matter-filled universe. Standard cosmological perturbation theory resolves the analogous Newtonian issue by using the *density contrast* in the fluctuation equation while the homogeneous density enters the background constraint; see [20, 22, 21]. This technique is a structural analogy, *not* a derivation of its applicability to a new QGD collective field.

On a comoving three-torus of coordinate volume V_c, separate $\rho = \bar{\rho} + \delta\rho$ with $\int_{V_c} \delta\rho \, \mathrm{d}^3x = 0$. Postulate only for the $\boldsymbol{k} \neq \mathbf{0}$ static-perturbation limit the Chapter 5 response law

$$\boxed{-Za^{-2}\nabla_{\boldsymbol{x}}^2 \delta q = g_c \delta\rho, \qquad \delta q_{\boldsymbol{k}} = \frac{g_c a^2}{Z|\boldsymbol{k}|^2}\delta\rho_{\boldsymbol{k}} \quad (\boldsymbol{k} \neq \mathbf{0}).} \tag{41}$$

The $\boldsymbol{k} = 0$ equation yields $0 = 0$: it fixes neither $\bar{q}(t)$ nor $H(t)$. On noncompact $\mathbb{R}^3$ the zero mode requires separately declared falloff, distributional and infinite-volume conditions. In either topology, one must not take the $\boldsymbol{k} \to 0$ limit of the inverse $1/|\boldsymbol{k}|^2$ and claim to have derived an autonomous homogeneous background law. If a physical homogeneous mode exists, its time-response kernel, initial data, energy, and effective-gravity coupling must be independently supplied.

Proposition 15.1 (Static inverse does not select a homogeneous history [D]|[C])**.** *With the mode splitting just described, the static $\boldsymbol{k} \neq 0$ Chapter 5 Green function constrains inhomogeneous potential fluctuations but makes no prediction of $\bar{q}(t)$ or $a(t)$. Moreover, the positive homogeneous static Hartree curvature K_0 tested in Chapter 6 cannot by itself determine the sign of a temporal kinetic coefficient, the position of retarded poles, or a cosmological instability.*

Proof. The periodic zero mode of a Laplacian vanishes identically; its inverse exists only on mean-zero functions. A static Hessian is one value/limit of a dynamical kernel, while distinct causal functions $K_R(\omega, 0)$ can have the same value $K_R(0, 0)$ and different pole spectra. For example, both $K_R = K_0 - Z_t(\omega + i0)^2$ and $K_R = K_0 + Z_t(\omega + i0)^2$ share $K_R(0, 0) = K_0$ but exhibit different formal pole structures for $K_0 Z_t > 0$. This algebraic comparison does not assign either example to the actual Hartree system. □

A homogeneous perturbation is also not automatically a gauge-invariant cosmological observable: it may change background parameters, slicing, or the meaning of the spatial average. Standard perturbation theory uses gauge-invariant variables and explicit matching to the separate-universe limit [21, 22]. QGD requires *both* the spacetime gauge quotient and Chapter 6's signed-sector $O(4, 4)$ representation quotient before counting independent physical modes.

16 Microscopic finite-temperature/real-time response: precise boundary protocol

16.1 What can be computed now, and what still requires a model

The Chapter 1 Dirac/scalar-contact action permits a homogeneous Hubbard–Stratonovich saddle $M = m_0 - s$. In a locally equilibrated finite-temperature/density state the formally renormalized, regulator-matched grand-potential density $\Omega_{\mathrm{ren}}(M; T, \mu)$ obeys

$$\boxed{\left.\frac{\partial\Omega_{\mathrm{ren}}}{\partial M}\right|_{M_*(T,\mu)} = 0, \qquad K_{MM}^{\mathrm{stat}} = \left.\frac{\partial^2\Omega_{\mathrm{ren}}}{\partial M^2}\right|_{M_*}.} \tag{42}$$

The static sign $K_{MM}^{\rm stat} > 0$ means local *thermodynamic* stability in that particular scalar channel. Chapter 6's benchmark value $K_0 = 0.2190085055 > 0$ was obtained for a named stationary regulated state; it is *not* a finite-temperature cosmic mode calculation. The homogeneous thermodynamics yields, with derivatives evaluated on the stationary saddle,

$$\boxed{p = -\Omega_{\rm ren}, \quad n = -\partial_\mu \Omega_{\rm ren}, \quad s_{\rm ent} = -\partial_T \Omega_{\rm ren}, \quad \epsilon = \Omega_{\rm ren} + T s_{\rm ent} + \mu n.} \tag{43}$$

These thermodynamic identities hold within a specified equilibrium ensemble and vacuum-subtraction scheme. They are not an emergent gravitational kinetic action. Local equilibrium further requires an independently assessed scattering/relaxation time, typically $H\tau_{\rm relax} \ll 1$ for a hydrodynamic description and $k_{\rm phys}\ell_{\rm mfp} \ll 1$ for long-wavelength gradients. In the opposite regime, a collisionless kinetic or full nonequilibrium evolution must be used. Expanding-background 2PI calculations provide a precedent for the latter [27, 28, 29].

For a source-sensitive dynamical mode, choose the quantum initial density matrix $\varrho(t_0)$, a regulated microscopic action and specified renormalization conditions. For physical composite observables $\widehat{O}_A$ (including the *full* stress tensor, not just $\bar{\psi}\psi$) define, in the spatially homogeneous background, the retarded two-time response

$$\boxed{\chi^R_{AB}(t,t';\boldsymbol{k}) = -i\,\Theta(t-t')\big\langle[\widehat{O}_A(t,\boldsymbol{k}), \widehat{O}_B(t',-\boldsymbol{k})]\big\rangle_{\varrho(t_0)}.} \tag{44}$$

Here $\hbar = 1$ and appropriate contact terms, composite-operator subtractions, time-dependent measures, and conserved stress vertices must be handled in the actual renormalized calculation. Translation invariance in *space* allows comoving $\boldsymbol{k}$, but there is generally no conserved Fourier frequency ω in FLRW. A time-local $K_R(\omega, \boldsymbol{k}; t)$ is only a controlled adiabatic/Wigner approximation, not an exact stationary correlator.

The candidate collective field φ^I must be specified as a physical renormalized observable or auxiliary variable of the microscopic model. On a background solution, a causal linearized sector equation has the schematic integral form

$$\boxed{\int_{t_0}^{t} \mathrm{d}t'\, \mathcal{K}^R_{IJ}(t,t';\boldsymbol{k})\, \delta\varphi^J(t',\boldsymbol{k}) = \delta J_I(t,\boldsymbol{k}), \qquad \boldsymbol{k} = 0 \text{ treated separately.}} \tag{45}$$

The "retarded Hessian" denotes the *physical, constrained in-in linear-response operator*, which need not coincide with the static thermodynamic curvature or a bare Euclidean functional Hessian. Its proper initial-value problem may include local equal-time terms and initial correlations. Inverting it must be done only on admissible physical perturbations after fixing gauge and sector redundancy. Its source vertices to density, pressure, anisotropic stress and photons must be *computed*; a massless density pole alone does not imply a universal Einstein-compatible gravitational interaction.

16.2 Comparison with established frameworks and physical boundaries

Framework	What it actually supplies	Requirement for QGD
Cosmological perturbations/Jeans subtraction	Mean density sources background; zero-mean density contrast solves the spatial Poisson equation; superhorizon modes need gauge/gradient control [20, 22].	Split $\boldsymbol{k} = 0$ from $\boldsymbol{k} \neq 0$ and justify any reference density subtraction from the microscopic/background dynamics, not just a choice of notation.
Semiclassical and stochastic gravity	Renormalized $\langle T_{\mu\nu}\rangle$ sources effective geometry; CTP gives causal response; stress fluctuations enter a *noise kernel*, not an extra arbitrary local metric slot [25, 26].	Use a specified initial state, covariant renormalization, retarded stress kernel, causal boundary data, and no double count via independent $\mathcal{Q}_{ab}$.

Framework	What it supplies	QGD translation
Nonequilibrium QFT / 2PI	Self-consistent contour evolution in expanding backgrounds can include scattering and memory; Φ-derivable approximations preserve some global conservation laws, but not all Ward identities [27, 28, 29].	Match a real $\Gamma_{\rm coll}$; test Ward identities, regulator dependence, initial transients and stability, rather than copying a scalar $\ddot{q}+3H\dot{q}$ equation.
Analogue/induced gravity	A collective normal mode may propagate in an effective Lorentzian metric, but kinematic geometry alone does not establish Einstein dynamics or universal coupling; induced terms require UV matching [30, 31].	Demonstrate common source coupling, healthy tensor modes, correct effective action and energy observables; do not identify a phonon or arbitrary sector gradient with gravity.

16.3 Explicit acceptance conditions for a cosmological collective mode

A candidate passes only if all applicable parts of the following *independent* tests are met; nothing in v1.2 asserts that the Chapter 1 model already passes them.

1. **State/renormalization:** choose $\varrho(t_0)$, admissible short-distance state (e.g. Hadamard/adiabatic conditions where appropriate), UV regulator and curvature counterterms; match the finite renormalized gravitational couplings. A FLRW Dirac $\langle T_{\mu\nu}\rangle_{\rm ren}$ is calculable but state dependent [24, 23].
2. **Causal initial-value problem:** use an in-in/retarded kernel with specified t_0, initial correlations and physically allowed boundary data. Reject exponentially growing *runaway artifacts* without interpreting them automatically as inflation; check timescales and validity of the derivative expansion [25].
3. **Symmetry and energy:** verify $\nabla_\mu\langle T^{\mu\nu}_{\rm total}\rangle = 0$ with the required local/contact Ward terms, renormalized pressure and microscopic Noether energy; enforce the Chapter 6 signed-orbit quotient before assigning new sector energies. Φ-derivable conservation is not by itself proof of all gauge Ward identities [29].
4. **Background/fluctuation split:** determine an independent $\boldsymbol{k}=0$ equation and admissible density mean; for $\boldsymbol{k}\neq 0$ verify the conditional Chapter 5 Newtonian/static limit and a causal long-wave continuation. Check that limits $\omega\to 0$ and $|\boldsymbol{k}|\to 0$ can be interchanged *only if proven.*
5. **Stability and universality:** after constraints, require nonnegative physical spectral weight in equilibrium where applicable, no uncontrolled ghost/tachyon/gradient instability, light-cone consistency, correct metric tensor modes and common stress/radiation vertices. A scalar gapless mode or mere rank $D_\sigma g = 10$ does not suffice.
6. **Cosmological closure and observability:** solve jointly for the renormalized source, $a(t)$ and physical collective observables; reproduce radiation/dust limits; calculate actual $H(z)$, $w(z)$ and perturbations before claiming inflation, dark energy or departures from GR. Chapter 6 Misner–Sharp/Noether energy is a consistency diagnostic, not a missing evolution equation.

Decision from the completed benchmark. The free Dirac kinetic sector gives a concrete homogeneous response and excludes acceleration *on its isolated conditional Einsteinian branch.* The static Chapter 5 $\boldsymbol{k}\neq 0$ response survives as a viable *conditional* perturbation model, but cannot be used as an equation for the cosmic zero mode. The existing positive Hartree scalar Hessian constrains a local static saddle, not a propagating long-range gravitational response. The first computation able to change this status is the *renormalized, retarded, source-resolved two-time stress/collective kernel* with explicit initial data and a demonstrated map to physical $[\Sigma]$. No unique off-shell $\mathcal{Q}_{ab}$ or independent vacuum energy was introduced.

17 Finite-temperature contact matter: a first interacting cosmological closure

Purpose and epistemic scope. This section changes no earlier theorem. It extends the *same* Chapter 1 four-fermion scalar-contact theory and Chapter 6 regulated Hartree benchmark to thermal equilibrium, then computes (i) an interacting matter equation of state; (ii) a *conditional* adiabatic FLRW history assuming the adopted Einstein equations; and (iii) a real-frequency, zero-spatial-momentum scalar response about a stationary thermal state. Equilibrium cosmology and collisionless retarded fluctuations are deliberately *distinct dynamical limits*. Neither prescribes an autonomous law for the full QGD natural invariants. We use $\hbar = c = k_B = 1$ throughout and denote fermionic quasiparticle mass by M, not gravitational mass.

17.1 Regulated grand potential, equilibrium gap, and thermodynamic stress

Take $\mathcal{L}_{\rm F1} = \bar{\psi}(i\gamma^\mu\partial_\mu - m_0)\psi + G_4(\bar{\psi}\psi)^2$, with a homogeneous Hartree auxiliary mass M, a fixed *physical* three-momentum cutoff Λ, and $\mu = 0$. For one Dirac species (two spin states, both particles and antiparticles), the precise continuation of the Chapter 6 prescription is

$$\boxed{\begin{aligned}\Omega(M,T) = \frac{(M-m_0)^2}{4G_4} - \frac{1}{\pi^2}\int_0^\Lambda dp\, p^2 E_p \\ - \frac{2T}{\pi^2}\int_0^\Lambda dp\, p^2 \ln(1+e^{-E_p/T}), \qquad E_p = \sqrt{p^2+M^2}.\end{aligned}} \tag{46}$$

This is a cutoff *matter* thermodynamic functional, not a new master-metric correction $-\lambda\mathcal{Q}_{ab}$. Its thermal factor counts fermion and antifermion occupations. With $n_p = (e^{E_p/T}+1)^{-1}$, the stationary mass obeys

$$\boxed{\frac{\partial\Omega}{\partial M} = \frac{M-m_0}{2G_4} - \frac{M}{\pi^2}\int_0^\Lambda \frac{p^2\,dp}{E_p}(1-2n_p) = 0.} \tag{47}$$

This is a *matter-sector* saddle equation; it does not select $a(t)$ without gravitational equations. Select the thermodynamically preferred regular stationary branch, subject to the validity of the cutoff/Hartree approximation. Define the fixed vacuum reference $\Omega_v = \Omega(M_v, 0)$, where M_v solves Eq. (47) at $T = 0$. The on-shell entropy, pressure and energy density are

$$\boxed{\begin{aligned} &s(T) = -\left.\frac{\partial\Omega}{\partial T}\right|_{M=M_*(T)} = \frac{2}{\pi^2}\int_0^\Lambda dp\, p^2\left[\ln(1+e^{-E_p/T}) + \frac{E_p}{T}n_p\right], \\ &p(T) = -\{\Omega[M_*(T),T] - \Omega_v\}, \\ &\epsilon(T) = \Omega[M_*(T),T] - \Omega_v + Ts(T).\end{aligned}} \tag{48}$$

Because $\Omega_M = 0$ on the saddle, the envelope theorem gives the *exact within-model* thermodynamic identity

$$\boxed{\frac{d\epsilon}{dT} = T\frac{ds}{dT}, \qquad \epsilon + p = Ts.} \tag{49}$$

The subtraction Ω_v is a *chosen* normalization; it does not determine nature's cosmological constant, and a sharp spatial cutoff does not replace curved-spacetime covariant renormalization. In the present homogeneous equilibrium approximation one adopts the perfect-fluid stress $T^\mu{}_\nu = \mathrm{diag}(-\epsilon, p, p, p)$; deriving the same result from a fully covariantly regulated FLRW effective action remains a separate test. Standard finite-T NJL Hartree gap and RPA procedures motivate the method but not any new gravitational claim [17, 18].

Proposition 17.1 (Conditional isentropic thermal cosmology [C]). *Suppose the selected thermal branch of Eq. (47) is in local equilibrium with negligible entropy production (requiring a microscopic relaxation time $\tau_{\rm relax}$ such that $H\tau_{\rm relax} \ll 1$), and suppose the flat Einstein–Friedmann equation with $\Lambda_{\rm cosm} = 0$ governs the matter stress in Eq. (48). Then for a selected reference (a_*, T_*) the expanding solution is obtained parametrically from*

$$\boxed{\begin{aligned} a(T) &= a_* \left[\frac{s(T_*)}{s(T)}\right]^{1/3}, \qquad H(T) = \sqrt{\frac{8\pi G_N}{3}\epsilon(T)}, \\ t(T) - t(T_i) &= \int_{T_i}^{T} \frac{d\ln a(T')/dT'}{H(T')}\, dT'. \end{aligned}} \tag{50}$$

These are a solvable conditional initial-value problem*, not a new gravity equation. Where $\epsilon > 0$, $p \geq 0$, acceleration is absent: $\ddot{a}/a = -(4\pi G_N/3)(\epsilon + 3p) \leq 0$.*

Proof. The fixed comoving entropy condition $d(sa^3)/dt = 0$ gives the first equality. Equation (49) and $\epsilon + p = Ts$ imply $\dot{\epsilon} = -3H(\epsilon + p)$ exactly. The adopted Friedmann constraint gives the second equality; $H = \dot{a}/a$ yields the time quadrature. The final inequality follows from the adopted Einstein acceleration equation, not from an independent QGD gravity law. □

17.2 Controlled numerical branch and its physical boundaries

Retain *exactly* the Chapter 6 benchmark $\Lambda = 4$, $m_0 = 0.12$, $G_4 = 0.5654255716$ in cutoff units. Numerical quadrature and gap solving reproduce $M_v = 0.800000000$ and $K_0(M_v, 0) = 0.2190085055$. Along this stationary branch, the computed values are:

T	$M(T)$	ϵ	p	p/ϵ	$K_T = \Omega_{MM}$
0.20	0.798522	0.00053	0.00009	0.17034	0.21761
0.40	0.766996	0.02152	0.00527	0.24481	0.20651
0.60	0.680957	0.12112	0.03587	0.29616	0.19910
0.80	0.567203	0.33154	0.11699	0.35287	0.21033

In contrast with the *free, collisionless fixed-mass* result of Section 14, the interacting vacuum-subtracted thermodynamic p/ϵ need not be bounded by 1/3. The two statements concern different model assumptions; $p/\epsilon > 1/3$ is not itself acceleration, which requires $\epsilon + 3p < 0$. All the displayed equilibrium branch points have positive ϵ and p, so they are decelerating if they alone source the adopted flat Einstein equations.

17.3 A real-frequency microscopic calculation of the homogeneous scalar kernel

An equilibrium Hessian is not automatically a causal kinetic law. To test the *actual homogeneous amplitude channel*, freeze a stationary Hartree thermal state at given (M, T) and treat a small oscillatory $\delta M(t)$ in the *collisionless one-loop/RPA* approximation. The Dirac Hamiltonian is $\mathcal{H}_{\boldsymbol{p}} = \boldsymbol{\alpha}\cdot\boldsymbol{p} + \beta M$ and the perturbing operator is $\beta\,\delta M$. Interband transitions have energy $2E_p$ and a scalar form factor proportional to p^2/E_p^2. The retarded inverse response, in the same cutoff and degeneracy convention as Eq. (46), is

$$\boxed{K_R(\omega, \mathbf{0}) = \frac{1}{2G_4} - \frac{4}{\pi^2}\int_0^{\Lambda} dp\, \frac{p^4(1 - 2n_p)}{E_p[4E_p^2 - (\omega + i0^+)^2]}.} \tag{51}$$

This is the homogeneous *amplitude*, not a full stress–stress or photon response, and only a stationary-state kernel: it is not yet a two-time $K_R(t, t'; \boldsymbol{k})$ on an expanding geometry. The analytic continuation

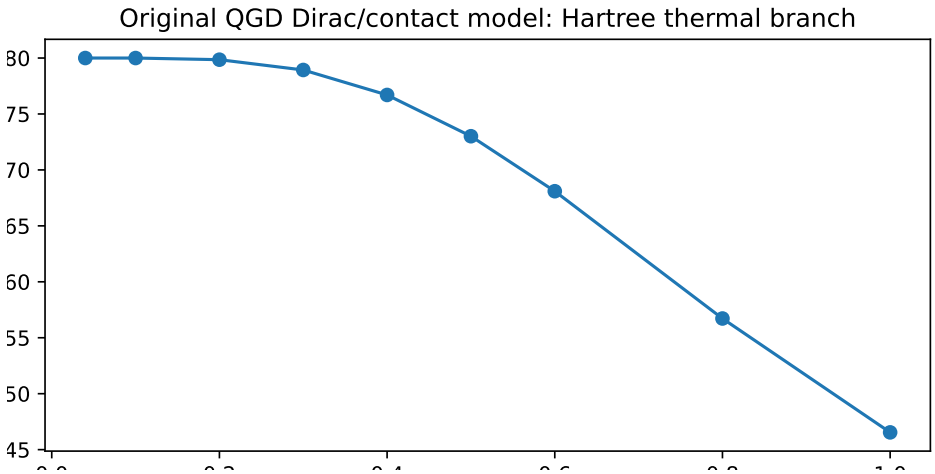


Figure 1: [C] The self-consistent Hartree quasiparticle mass decreases as the thermal state becomes hotter. The numerical parameters are the Chapter 6 benchmark, not cosmological best-fit parameters.

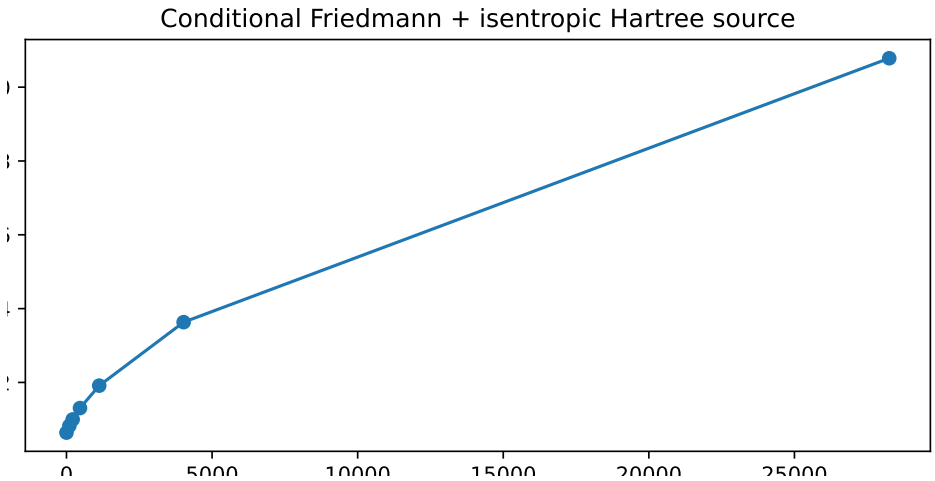


Figure 2: [C] A computed but *illustrative* isentropic FLRW scale factor. We use $a(T = 0.5) = 1$ and choose $8\pi G_N = 10^{-4}$ in cutoff units solely to set a slow-expansion time scale; G_N is *not* derived or matched. The universe is decelerating along this branch. Collision rates and the finite-temperature covariant stress have not been computed.

is retarded; the particle–antiparticle continuum begins at $|\omega| = 2M$. The corresponding equilibrium second derivative contains an additional occupation-relaxation term:

$$\boxed{K_T \equiv \Omega_{MM} = K_R(0,\mathbf{0}) - \frac{2M^2}{\pi^2 T}\int_0^\Lambda dp\,\frac{p^2}{E_p^2} n_p(1-n_p).} \tag{52}$$

The distinction records the noncommuting collisionless real-time and equilibrated static limits. It is not a new collective force. At $T = 0$ the thermal relaxation term vanishes and the Chapter 6 Hessian K_0 is recovered.

Proposition 17.2 (Exact subthreshold scalar-pole exclusion for $m_0 > 0$ [D])**.** *Assume the regulated positive-mass gap equation* (47) *holds with $M > 0$, $m_0 > 0$, $G_4 > 0$, $T \geq 0$, $\mu = 0$, and a stationary collisionless thermal occupation $0 \leq n_p \leq \frac{1}{2}$. For every real frequency $|\omega| \leq 2M$, the one-loop/RPA scalar inverse response* (51) *satisfies*

$$\boxed{\begin{aligned} K_R(\omega,\mathbf{0}) &= \frac{m_0}{2G_4M} + \frac{4M^2-\omega^2}{\pi^2}\int_0^\Lambda dp\,\frac{p^2(1-2n_p)}{E_p(4E_p^2-\omega^2)} \\ &\geq \frac{m_0}{2G_4M} > 0. \end{aligned}} \tag{53}$$

At $|\omega| = 2M$ the integrand is taken in the finite threshold limit, and equality holds. Therefore this channel has no real isolated subthreshold pole *in the stated collisionless homogeneous approximation. This does not exclude continuum resonances, complex poles, nonzero-momentum modes, critical $m_0 = 0$ limits, other fermionic/stress channels or a different phase.*

Proof. By Eq. (47), $1/(2G_4) - (1/\pi^2)\int p^2(1-2n_p)/E_p\,dp = m_0/(2G_4M)$. Substitute this into Eq. (51) and combine the integrands using $E_p^2 = p^2 + M^2$. For $|\omega| < 2M$, both the denominator $4E_p^2 - \omega^2$ and $(4M^2 - \omega^2)$ are nonnegative; $1 - 2n_p \geq 0$. Dominated threshold evaluation is finite with the given cutoff. This proves the inequality and, since the scalar response pole requires a zero of K_R, its absence in the specified interval. □

Numerical independent check. The Chapter 6 zero-temperature scalar Hessian $K_T = K_R(0) = 0.2190085055$ is reproduced. At $T = 0.4$, the equilibrium Hessian is $K_T = 0.206513$, the collisionless real-time static-limit value is $K_R(0) = 0.216520$, and the exact pair threshold is $K_R(2M) = 0.138351 > 0$. At $T = 0.8$, these values are 0.210328, 0.227772, and 0.187084. The symbolic threshold identity and these values were verified using quadrature independent of a fitted cosmological expansion history. This is consistent with standard NJL scalar-channel RPA threshold structure [17, 18].

17.4 What has and has not been predicted

The calculation is a genuine microscopic *matter* result: the chosen G_4, m_0, Λ give a stationary $M(T)$, $\epsilon(T)$, $p(T)$ and a subthreshold scalar inverse-response bound. Given *extra* equilibrium, entropy, Friedmann and coupling assumptions, it also produces the explicit conditional expansion $a(t)$. But the scalar mode in this tested channel *cannot* be promoted to QGD's universal gravity merely because it has a thermodynamic energy and a response kernel. In fact the subthreshold no-pole result makes this particular identification less plausible. The bare scalar auxiliary field couples to $\bar\psi\psi$, not universally to $T_{\mu\nu}$ or Maxwell radiation. The regulator is not covariant, quantum state perturbations are not solved through FLRW in-in evolution, collision rates are not known, the physical [Σ] map has not been extracted, and no independent QGD relation selecting G_N was computed. The $\lambda\mathcal{Q}_{ab}$ slot remains unmatched and is not activated here.

Required next test: derive source-resolved $\chi^R_{AB}(t,t';\boldsymbol{k})$ for energy density, pressure, shear and photon sources in a regulated in-in/2PI framework; include Ward/contact terms and the initial

quantum state, test the $\boldsymbol{k} = 0$ and $\boldsymbol{k} \neq 0$ limits independently, quotient the unphysical signed-sector orbit, and verify physical poles and universal vertices before inferring a gravitational origin [19, 28, 29]. The current section is an explicit *positive thermodynamic calculation and negative subthreshold scalar-pole test*, not a proof of inflation, dark energy or fundamental microscopic cosmological gravity.

18 Quantum vacuum, covariant curvature matching, and the cosmological constant (v1.4)

Research question and scope. The historical cosmology asked whether a gravitational vacuum or a quantum sector could explain cosmic acceleration. We now test that question using the *renormalized QGD–136 effective action* from Chapter 4, not only the classical QGD–120 action, while keeping the complete founding master metric (2). In the working metric-only branch $\lambda\mathcal{Q}_{ab} = 0$, quantum effects enter the matter functional $\Gamma_{\mathrm{m},ren}$ and the resulting metric response $\delta g^{\mathrm{quantum}}$; they are not counted again as an independent $\mathcal{Q}_{ab}$. These calculations *do not derive* the Einstein–Hilbert term from quantum matter, and all curvature expansions below concern the adopted background metric.

18.1 The action relevant to cosmological quantum backreaction

In $c = \hbar = 1$ units, the current controlled Chapter 4 QGD–136 action, on a regular signed-sector branch, is

$$\boxed{\begin{aligned}\Gamma^{\mathrm{ren}}_{\mathrm{QGD}} = &\int \mathrm{d}^4x\sqrt{-g(\sigma)}\,\frac{R(g) - 2\Lambda_{\mathrm{ren}}}{16\pi G_N} + S_{\mathrm{GHY}}[g] + S^{\mathrm{cl}}_{\mathrm{F1}}[g,\psi_c,A_c] \\ &+ \hbar\,\Gamma^{(1)}_{\mathrm{m,ren}}[g,\psi_c,A_c;\varrho,\mu] + O(\hbar^2), \qquad g = g[\sigma;T], \quad \lambda\mathcal{Q} = 0.\end{aligned}} \tag{54}$$

Its *microscopic* loop functional and state are not already an autonomously derived gravitational collective action. Renormalization of the matter determinant requires the same gravitational couplings $\Lambda_{\mathrm{ren}}, G_N$ and curvature-squared coefficients that appear in the background equations. The reference coframe is retained; the signed-lift quotient prevents any sector-dependent arbitrary vacuum kinetic energy from being added without a separate microscopic observable.

18.2 Exact vacuum-shift ambiguity in the existing three-momentum cutoff model

For the same regulated scalar-contact Hartree model of Sec. 17, let $\Omega_v = \Omega(M_v, 0)$ denote a stationary vacuum value. With constant M_v and an idealized Lorentz-invariant, covariantly normalized vacuum, the volume term produces

$$\boxed{T^{\mathrm{vac}}_{\mu\nu} = -\epsilon_v g_{\mu\nu}, \qquad p_v = -\epsilon_v, \qquad \Lambda_{\mathrm{eff}} = \Lambda_{\mathrm{ren}} + 8\pi G_N \epsilon_v.} \tag{55}$$

This describes the *form* of a vacuum contribution, not its observable magnitude. The Chapter 7 three-momentum-cutoff grand potential alone does not give a covariant FLRW stress tensor; Eq. (55) applies only after an appropriate covariant gravitational matching.

Proposition 18.1 (Vacuum-shift non-identifiability [D] within the current effective action)**.** *If the matter grand potential is replaced by $\Omega'(M,T) = \Omega(M,T) + C$ for a field- and temperature-independent constant C, its saddle $M_*(T)$, equilibrium entropy, and vacuum-subtracted $\epsilon(T), p(T)$ of Eq. (48) are unchanged. A compensating gravitational parameter shift*

$$\boxed{\epsilon'_v = \epsilon_v + C, \qquad \Lambda'_{\mathrm{ren}} = \Lambda_{\mathrm{ren}} - 8\pi G_N C, \qquad \Lambda'_{\mathrm{eff}} = \Lambda_{\mathrm{eff}}} \tag{56}$$

leaves the cosmological metric equations unchanged at this order. Thus the existing finite-cutoff matter calculation, by itself, cannot predict the absolute observed $\Lambda_{\rm eff}$.

Proof. The constant disappears upon taking the M or T derivative, while $\Omega'(M,T) - \Omega'_v = \Omega(M,T) - \Omega_v$. In the effective gravitational equations, the variation of a constant matter vacuum contribution is proportional to the metric, and the displayed Λ-countershift removes it. The result is an algebraic renormalization ambiguity, not a proof of the absence of physical vacuum stress. □

Numerical check on the unchanged Chapter 6/7 branch. With $\Lambda_{3D} = 4$, $m_0 = 0.12$, $G_4^{3D} = 0.5654255716$, $M_v = 0.8$, direct quadrature gives

$$\boxed{\Omega_v = -6.5287908865, \quad \Omega(m_0, 0) = -6.4903814795, \quad \Omega_v - \Omega(m_0, 0) = -0.0384094070.} \tag{57}$$

All numbers have the model's cutoff units (energy density in cutoff units to the fourth power). The *difference* is invariant under adding a common C to the same regulated potential; $M = m_0$ is a comparison configuration, not necessarily another stationary phase. The un-subtracted Ω_v may not be called a measured vacuum energy or a prediction of an AdS universe. In particular, adding the arbitrary C reverses its apparent sign without changing the flat-space gap or thermal calculations. Matching a physical gravitational renormalization condition remains necessary [10, 11].

18.3 A covariant one-loop curvature response from the same Dirac interaction

The flat-space sharp three-momentum cutoff in Eq. (46) is not a generally covariant regulator. To examine curvature, introduce a *separate proper-time scheme* and rematch its bare coupling instead of combining unmatched coefficients. In a slowly varying Euclidean geometry, suppress background Maxwell fields and take a constant auxiliary mass M; the parity-even Dirac modulus gives

$$\boxed{\Gamma^{(1)}_{E,f} = -\ln\det(\not{D}_E + M) = -\frac{1}{2}\,\mathrm{Tr}\ln\mathcal{P}_M + (\text{phase}), \qquad \mathcal{P}_M = -\nabla^2 + \frac{R}{4} + M^2.} \tag{58}$$

The sign of the Euclidean spinor Laplacian is chosen so that $\mathcal{P}_M$ is positive in the flat massive limit. By the Lichnerowicz identity and the heat-kernel coefficient $a_1 = R/6 - E$ for an operator $-\nabla^2 + E$, one full four-component Dirac species obeys

$$\boxed{\mathrm{Tr}\, e^{-s\mathcal{P}_M} \sim \frac{1}{(4\pi s)^2}\int \mathrm{d}^4x\sqrt{g_E}\, e^{-sM^2}\left(4 - \frac{sR_E}{3} + O(s^2\mathcal{R}^2,\, s^2\nabla^2 R)\right).} \tag{59}$$

For one explicit ultraviolet proper-time cutoff $s \geq \Lambda_{\rm PT}^{-2}$, the corresponding *unrenormalized local* Euclidean density contains

$$\boxed{\begin{aligned}\mathcal{V}_{E,\rm loc}(M, R_E) &= \frac{(M-m_0)^2}{4G_4^{\rm PT}} + \frac{I_2(M)}{8\pi^2} + C_R^E(M)R_E + O(R_E^2, \nabla M),\\ I_j(M) &= \int_{\Lambda_{\rm PT}^{-2}}^{\infty}\mathrm{d}s\, s^{-j-1}e^{-M^2 s}, \qquad C_R^E(M) = -\frac{I_1(M)}{96\pi^2}.\end{aligned}} \tag{60}$$

The coefficient C_R^E renormalizes the Einstein–Hilbert normalization after signature continuation and matching; its raw value and separate sign are *not* a parameter-free physical shift of G_N. Direct a_2 curvature-squared terms, nonlocal/state terms, derivatives of M, photon and auxiliary fluctuations, gravitational counterterms, and possible determinant phases are not computed by Eq. (60). This is a legitimate one-loop *local* curvature expansion, not a full FLRW causal effective action [9, 12, 11].

Proposition 18.2 (Adiabatic local curvature response of the Hartree saddle [D] within the proper-time scheme)**.** *Let* $M_v > 0$ *solve the* proper-time-scheme *flat stationary gap equation and*

have positive $K_v = \partial_M^2 \mathcal{V}_{E,\mathrm{loc}}(M_v, 0) > 0$. *Define* $I_0(M) = \int_{\Lambda_{\mathrm{PT}}^{-2}}^{\infty} \mathrm{d}s\, s^{-1} e^{-M^2 s}$. *For weak, sufficiently slowly varying scalar curvature* R_E, *the local stationary mass shifts by*

$$\boxed{\begin{aligned} K_v &= \frac{m_0}{2G_4^{\mathrm{PT}} M_v} + \frac{M_v^2 I_0(M_v)}{2\pi^2} > 0, \\ M_*(R_E) &= M_v - \frac{C_R^{E\,\prime}(M_v)}{K_v} R_E + O(R_E^2), \\ C_R^{E\,\prime}(M_v) &= \frac{M_v I_0(M_v)}{48\pi^2} > 0. \end{aligned}} \tag{61}$$

Eliminating the auxiliary mass locally and adiabatically *adds the curvature-squared contribution*

$$\boxed{\Delta\mathcal{V}_{E,\text{on-shell}}^{(R^2)} = -\frac{[C_R^{E\,\prime}(M_v)]^2}{2K_v} R_E^2.} \tag{62}$$

This is only the condensate-relaxation part of the R^2 *coefficient; it must* not *be identified with the complete renormalized, dynamical* R^2 *coupling.*

Proof. The flat gap equation is $(M_v - m_0)/(2G_4^{\mathrm{PT}}) = M_v I_1(M_v)/(4\pi^2)$. Since $I_1' = -2MI_0$, differentiation and gap substitution give the first equation. The implicit-function theorem applied to $\partial_M \mathcal{V}_{E,\mathrm{loc}}(M, R_E) = 0$ gives the linear response. Substituting $M = M_v - C_R' R_E / K_v$ into the quadratic Taylor expansion $\frac{1}{2}K_v(M - M_v)^2 + C_R' R_E (M - M_v)$ gives Eq. (62). □

Separate covariant-scheme numerical benchmark. Set $\Lambda_{\mathrm{PT}} = 4$, $m_0 = 0.12$ and *rematch* G_4^{PT} to obtain $M_v = 0.8$. This gives

$$\boxed{\begin{gathered} G_4^{\mathrm{PT}} = 1.2285854552, \quad K_v^{\mathrm{PT}} = 0.1479798341, \\ C_R^E(M_v) = -0.0144135951, \quad \left.\frac{\mathrm{d}M_*}{\mathrm{d}R_E}\right|_0 = -0.0305975036, \\ -\frac{(C_R^{E\,\prime})^2}{2K_v} = -0.0000692699. \end{gathered}} \tag{63}$$

These are cutoff-model numbers, *not observed curvature coefficients.* In particular, $G_4^{\mathrm{PT}} \neq G_4^{\mathrm{3D}}$ and $K_v^{\mathrm{PT}} \neq K_0^{\mathrm{3D}}$; using the Chapter 6 $K_0 = 0.2190085055$ with C_R^E here would be an invalid mixed-scheme calculation. The appended verification script independently solves the proper-time gap at 33 curvature values and checks the perturbative derivative.

18.4 Does an induced curvature term select accelerated expansion?

The answer is *not by itself.* A local one-loop R coefficient only renormalizes the effective gravitational normalization after matching. An R^2 coefficient can affect *time-dependent* homogeneous expansion but does not automatically generate a nonzero constant-curvature de Sitter vacuum.

Proposition 18.3 (No nonzero de Sitter curvature from pure $R+\alpha R^2$ when $\Lambda_{\mathrm{eff}} = 0$ [C])**.** *Consider the* adopted *four-dimensional vacuum metric action* $S \propto \int \sqrt{-g}\, f(R)$ *with* $f(R) = R - 2\Lambda_{\mathrm{eff}} + \alpha R^2$ *and constant* α. *On an exact constant-curvature vacuum the metric trace equation is*

$$\boxed{Rf_R - 2f = 0 \quad \Longleftrightarrow \quad R = 4\Lambda_{\mathrm{eff}}.} \tag{64}$$

Consequently, if $\Lambda_{\mathrm{eff}} = 0$, *no nonzero constant-curvature de Sitter branch follows from the* R^2 *term alone. For spatially flat de Sitter,* $R = 12H^2$, *so* $H^2 = \Lambda_{\mathrm{eff}}/3$ *remains necessary at this level.*

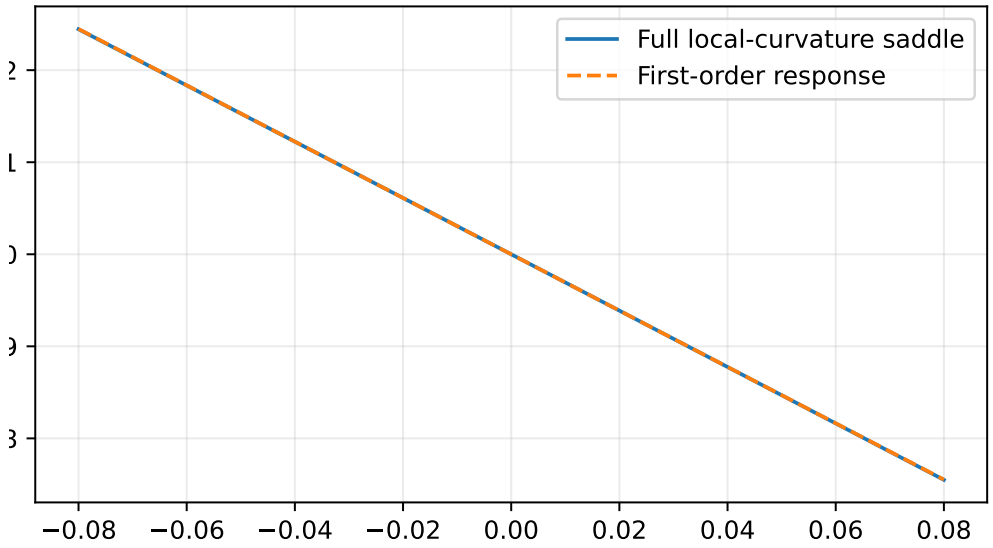


Figure 3: Local Euclidean Hartree response to weak scalar curvature in one *rematched proper-time scheme*. The dashed first-order prediction follows Eq. (61); the full curve solves the truncated curvature-dependent gap equation. This figure is a regulator-model benchmark, not a cosmological $M(t)$ simulation.

Proof. The four-dimensional vacuum $f(R)$ trace equation is $3\Box f_R + Rf_R - 2f = 0$. At constant R, the box term vanishes and $Rf_R - 2f = R(1 + 2\alpha R) - 2(R - 2\Lambda_{\text{eff}} + \alpha R^2) = -R + 4\Lambda_{\text{eff}}$. □

This theorem does *not* exclude slowly evolving R^2-driven inflation, which requires the complete higher-curvature dynamics, stability, appropriate initial data and a physical renormalized coefficient. Nor does it cover more general $f(R)$, nonlocal stress, state-dependent backreaction, or a genuinely derived additional collective mode. On the homogeneous FLRW background the Weyl tensor is zero, but the omission of its variations from perturbations would be unjustified.

18.5 Physical decision: which calculation could genuinely fix Λ?

1. **Independent vacuum matching:** supply a renormalization/microscopic matching condition that fixes the *sum* of the cosmological counterterm and the quantum volume term. Merely choosing $\Omega_v = 0$ is a convention, not a cosmological prediction.

2. **Covariant full interacting action:** include the fermion auxiliary field and its derivatives, Maxwell fields, state dependence and a covariant prescription for gravitational counterterms. The Hartree scalar potential alone does not fix the physical zero mode.

3. **Causal backreaction:** on FLRW derive the in–in $\langle T_{\mu\nu}\rangle_{\text{ren}}$, including initial state and any memory kernel, verifying conservation and positivity. An in–out heat-kernel density cannot replace an expanding-universe initial-value calculation.

4. **Microscopic-to-natural-invariant map:** calculate physical source vertices and the collective Hessian; decide whether $[\Sigma]$ is selected by quantum matter. The local geometric representation and $O(4,4)$ redundant modes supply no extra vacuum energy.

5. **Actual tests:** only after the above may a state-independent Λ_{eff}, running effective terms, $H(z)$ and perturbation observables be claimed as independently determined. Comparing with observed H_0 may calibrate, but cannot be called predicting Λ.

Research verdict. The Chapter 1 contact quantum vacuum supplies a well-defined *regulated relative* matter potential and a calculable *local bare curvature response* after a separate covariant

rematching. Neither determines a parameter-free cosmological constant. The exact $R + \alpha R^2$ trace result eliminates the simplest claim that a loop-induced R^2 correction automatically selects a nonzero de Sitter expansion. The next genuinely discriminating step is the *finite-momentum, full interacting* renormalized state-sensitive *causal* stress tensor with independently fixed gravitational counterterms, followed by microscopic collective source matching.

19 A first microscopic stress-energy response calculation (v1.6)

Research question. Does the actual Chapter 1 microscopic Dirac/contact matter sector generate a physical, source-visible long-range *tensor* excitation? Unlike a hypothetical free σ_t oscillator, this question can be tested against the conserved stress-energy tensor. The following is a precise, *restricted* calculation: flat, stationary, isotropic $\mu = 0$ equilibrium; a Hartree quasiparticle mass $M = M_*(T) > 0$ from Section 17; leading Dirac one-loop bubble; and the same rotationally symmetric three-momentum regulator $|\mathbf{p}| < \Lambda_{3D}$ used to define that benchmark. The full stress tensor of the interacting contact/Maxwell theory and its diffeomorphism Ward identities must ultimately be obtained by varying its *complete* renormalized metric-dependent action. Our one-loop result does not establish those missing ingredients.

19.1 Homogeneous energy: a conservation theorem, not a propagating mode

Proposition 19.1 (Stationary homogeneous energy commutator [D])**.** *Let H_{micro} be a time-independent, spatially translation-invariant Hamiltonian in a periodic volume V, with a stationary quantum state $[\varrho, H_{\text{micro}}] = 0$. When $\widehat{T}^{00}_{\mathbf{k}=0} = H_{\text{micro}}$, its non-contact retarded commutator is zero at every frequency:*

$$\boxed{C_R^{00,00}(\omega, \mathbf{0}) = -i\int_0^\infty \mathrm{d}t\, e^{i\omega t} \left\langle \left[H_{\text{micro}}(t), H_{\text{micro}}(0)\right]\right\rangle /V = 0.} \tag{65}$$

Moreover, for any time-independent operator B whose thermal expectation is stationary, $\langle [H_{\text{micro}}, B(t)]\rangle = 0$, so the homogeneous mixed energy–B retarded commutator vanishes as well.

Proof. $H_{\text{micro}}(t) = H_{\text{micro}}$ and $[H_{\text{micro}}, H_{\text{micro}}] = 0$. For the mixed case, $\mathrm{d}\langle B(t)\rangle/\mathrm{d}t = i\langle [H_{\text{micro}}, B(t)]\rangle = 0$. The finite-volume normalization is immaterial to the zero. This proof concerns the *commutator piece*; a metric variation of $T^{\mu\nu}$ also produces Ward-required local contact/seagull terms, and an equilibrium change of temperature changes the state rather than this fixed-state retarded commutator. □

Remark 19.2 (Cosmological and low-momentum boundary). This theorem does *not* assert that matter energy is conserved independently in a time-dependent FLRW background; there, $\dot\epsilon + 3H(\epsilon + p) = 0$ describes expansion work and the microscopic Hamiltonian need not be stationary. It also does not imply zero thermal specific heat. The limits $\omega \to 0$ and $|\mathbf{k}| \to 0$ need not commute. At finite momentum the Ward identity is schematically $\omega C_R^{0\nu,\rho\sigma} - k_i C_R^{i\nu,\rho\sigma} = \mathcal{C}^{\nu,\rho\sigma}$ with local contact terms $\mathcal{C}$ fixed by the full metric variation. See Refs. [15, 13].

19.2 Tensor response: exact one-loop pair continuum

The symmetric Belinfante stress operator of the constant-mass Dirac quasiparticle theory has the spatial off-diagonal component

$$T^{xy} = \frac{i}{4}\bar\psi\big(\gamma^x \overleftrightarrow{\partial}^y + \gamma^y \overleftrightarrow{\partial}^x\big)\psi, \qquad E_p = \sqrt{\mathbf{p}^2 + M^2}. \tag{66}$$

Define the *pair-creation continuum only* by the thermal commutator convention $\rho_{xyxy}(\omega,\mathbf{0}) = \int \mathrm{d}t\,\mathrm{d}^3x\, e^{i\omega t}\langle[T^{xy}(t,\mathbf{x}),T^{xy}(0,\mathbf{0})]\rangle$, for positive ω. In the particle–antiparticle center-of-mass frame the two-particle vertex is

$$\mathcal{V}_{xy}(\mathbf{p}) = \frac{1}{2}\big(p_y\gamma^x + p_x\gamma^y\big), \qquad \mathscr{M}_{xy} = \bar{u}(E_p,\mathbf{p})\mathcal{V}_{xy}(\mathbf{p})v(E_p,-\mathbf{p}). \tag{67}$$

Proposition 19.3 (Free Hartree-quasiparticle shear spectral continuum [D] | [C])**.** *For one full four-component Dirac species at zero chemical potential in the stated leading quasiparticle approximation, the positive-frequency pair-production contribution is*

$$\boxed{\rho^{(\mathrm{pair})}_{xyxy}(\omega,\mathbf{0}) = \frac{\omega^4}{160\pi}\left(1-\frac{4M^2}{\omega^2}\right)^{3/2}\left(1+\frac{8M^2}{3\omega^2}\right)\tanh\Big(\frac{\omega}{4T}\Big), \quad 2M<\omega<2\sqrt{\Lambda_{3D}^2+M^2},} \tag{68}$$

with $\rho^{(\mathrm{pair})}_{xyxy}=0$ for $0<\omega<2M$ and outside the regulator's quasiparticle band. Set the tanh *factor to* 1 *at $T=0$. The result describes an onset of a* continuum*, not an isolated subthreshold tensor pole. It does not classify transport contributions localized near $\omega=0$ or higher-order interaction corrections.*

Proof. Use the Dirac completeness relations and $\eta=\mathrm{diag}(+,-,-,-)$. Writing $\not{p} = E_p\gamma^0 - \mathbf{p}\cdot\boldsymbol{\gamma}$ and $\not{q} = E_p\gamma^0 + \mathbf{p}\cdot\boldsymbol{\gamma}$, the positive spin sum is

$$\boxed{\sum_{s,r}|\mathscr{M}_{xy}|^2 = \mathrm{tr}\big[(\not{p}+M)\mathcal{V}_{xy}(\not{q}-M)\mathcal{V}_{xy}\big] = 2\left[E_p^2(p_x^2+p_y^2) - 4p_x^2p_y^2\right].} \tag{69}$$

Rotational averaging gives $\langle p_x^2+p_y^2\rangle = 2p^2/3$ and $\langle p_x^2p_y^2\rangle = p^4/15$, so $\overline{|\mathscr{M}_{xy}|^2} = \frac{4}{3}E_p^2p^2 - \frac{8}{15}p^4$. With the stated spectral convention and relativistic pair normalization, the cut is

$$\rho^{(\mathrm{pair})}_{xyxy}(\omega,\mathbf{0}) = 2\pi\int_{|\mathbf{p}|<\Lambda_{3D}}\frac{\mathrm{d}^3p}{(2\pi)^3}\,\frac{\overline{|\mathscr{M}_{xy}|^2}}{4E_p^2}\big[1-2n_F(E_p)\big]\delta(\omega-2E_p), \tag{70}$$

where $n_F(E)=(e^{E/T}+1)^{-1}$. The radial delta integral gives $\rho = p\overline{|\mathscr{M}_{xy}|^2}\tanh(E_p/(2T))/(8\pi E_p)$ with $p=\sqrt{\omega^2/4-M^2}$, which reduces to Eq. (68). Independently setting $M=0$ gives $\rho = \omega^4\tanh(\omega/(4T))/(160\pi)$ inside the cutoff band, fixing the normalization under our explicit convention. This leading free-loop calculation follows standard finite-temperature shear-stress spectral/Kubo methods [14, 13]. □

19.3 Why the scalar Hartree pole cannot simply be relabelled a graviton

Let $\mathcal{O}_s=\bar{\psi}\psi$ be the scalar collective operator used in the Chapter 7 v1.3 Hartree/RPA calculation. In the stationary isotropic state, a spatial rotation by $\pi/2$ about the z axis sends $T^{xy}\mapsto -T^{xy}$ while $\mathcal{O}_s\mapsto\mathcal{O}_s$. Rotational invariance consequently implies

$$\boxed{\chi^R_{xy,s}(\omega,\mathbf{0})=0, \qquad \chi^R_{s,xy}(\omega,\mathbf{0})=0.} \tag{71}$$

At $\mathbf{k}\parallel\hat{z}$, the helicity-two transverse-traceless response also cannot mix linearly with a scalar source in a rotationally invariant medium. Thus the *separable scalar-channel RPA correction* to the tensor response is zero:

$$\chi^{\mathrm{RPA}}_{xy,xy} = \chi^{(0)}_{xy,xy} + \chi^{(0)}_{xy,s}\left[K_s^R\right]^{-1}\chi^{(0)}_{s,xy} = \chi^{(0)}_{xy,xy} \qquad \text{(only the scalar-ladder approximation).} \tag{72}$$

This is [D] under isotropy and the stated channel decomposition. It does *not* mean four-fermion interactions have no effect on the true tensor response: stress-vertex corrections, collisions, counterterms, the full two-particle kernel and Maxwell contributions are not contained in this scalar ladder. Nor does the absence of an isolated tensor pole at $\mathbf{k}=0$ forbid hydrodynamic sound or shear modes at small but finite $\mathbf{k}$; their order-of-limits and branch-cut structures are distinct [15, 16].

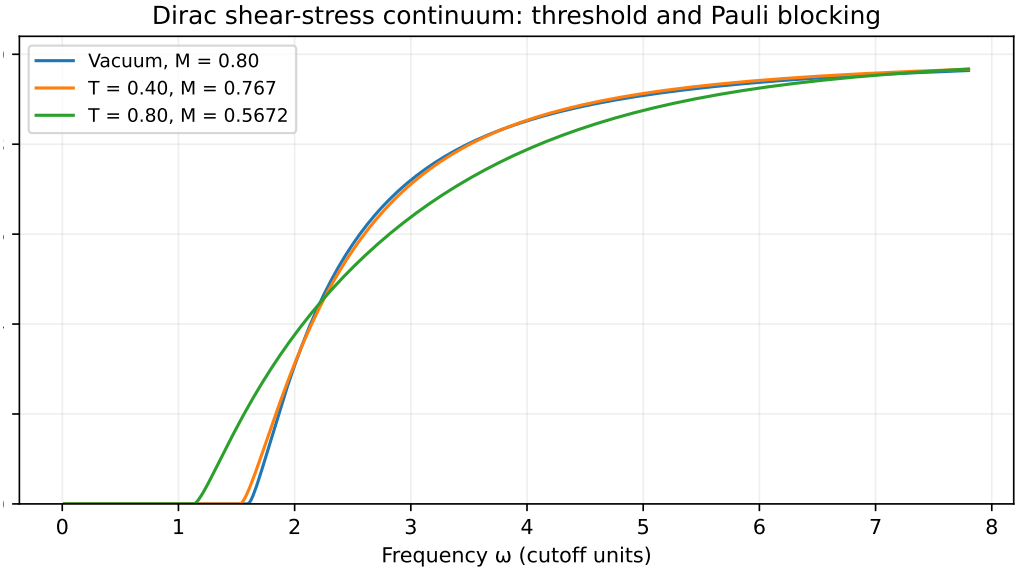


Figure 4: Calculated positive-frequency pair continuum, displayed as a ratio to the massless $T = 0$ $\omega^4/(160\pi)$ baseline (one Dirac species, three-momentum cutoff $\Lambda_{3D} = 4$). The $T > 0$ masses use the Chapter 7 v1.3 thermal Hartree solutions, rounded for the illustrative plot. A sharp threshold is not a bound-state pole; collisionless zero-frequency transport contributions are not plotted.

19.4 QGD interpretation and the remaining true gravitational calculation

The Chapter 5 universal-gravity criterion calls for a source-visible response with a healthy $|\mathbf{k}|^{-2}$ static limit, consistent $\omega(\mathbf{k})$ dynamics, coupling to energy, pressure, radiation and all relevant species, and an independently justified metric map. Equation (68) supplies *neither* a tensor bound-state pole nor any proof of a universal gravitational mode. More importantly, no finite-$\mathbf{k}$ tensor form factor has yet been calculated here; one cannot infer its full static limit from a single $\mathbf{k} = 0$ frequency cut. Chapter 6's exact $O(4,4)$ representation redundancies remain a separate geometric constraint: a mode in a microscopic stress correlator is not automatically a nonredundant $\sigma^{(s)}$ excitation.

The present tensor calculation is a local, approximately stationary tangent-space benchmark, *not* a full nonstationary FLRW in–in answer. Its equilibrium use requires microscopic time scales to be short compared with H^{-1} and the evolution time of $M_*(T)$; otherwise the two-time kernels $C_R(t, t'; \mathbf{k})$ must be computed directly. Moreover, a hard spatial cutoff does not define a generally covariant gravitational regularization at finite momentum. The next decisive calculation must: (i) derive the full interacting metric-variation stress vertex with its contact terms and renormalization; (ii) compute scalar, vector and tensor form factors at small finite $\mathbf{k}$ in a conserving approximation; (iii) include real collision/transport and electromagnetic matter vertices; (iv) test real-time poles, positivity and universality; and (v) only then match a physical quantum collective state to the complete founding metric with the historical $-\lambda\mathcal{Q}_{ab}$ not independently double-counted.

v1.6 statement	**Proof status and boundary**
$C_R^{00,00}(\omega, \mathbf{0}) = 0$	[D] for stationary, closed translation-invariant states, non-contact commutator only; not FLRW time evolution.
Exact $\rho_{xyxy}^{(\mathrm{pair})}$	[D] for one constant-mass free Dirac bubble, [C] for Hartree quasiparticle matching; regulator band and transport excluded.
Scalar–TT decoupling	[D] for an isotropic scalar channel in the separable RPA, not full contact-model tensor dynamics.
Long-range, universal gravitational mode	[O]: no physical source-resolved finite-momentum pole/metric matching obtained.

20 Historical-claim reconciliation ledger

Historical idea / equation	What the six chapters establish	Chapter 7 disposition
One $\sigma_t(t)$ uniquely fixes isotropic FLRW	An individual covector can respect rotational symmetry, but a purely temporal deformation cannot create $a^2\delta_{ij}$ with fixed flat coframe	Replace with exact three-spatial-pair lift; add time pair for rank ten
$\Box\sigma_\mu = Q_\mu + G_\mu + T_\mu + \ell_Q^2\Box^2\sigma_\mu$	Action-derived full map is Eq. (3); old wave equation unmatched	Retain only as archival hypothesis; no independent fourth-order dynamics asserted
Historical four roots $0, -3H, \pm 1/\ell_Q$	Fourth-order scalar operators need precise covariant definition; covectors differ	Exhibit correct scalar trial identity and Hubble-shifted roots; no physical gravity claim
Growing mode drives/saturates inflation	No nonlinear saturation equation or stable physical Hamiltonian derived	Open test with action, backreaction, reheating, perturbations
Kinetic-only $w = -1$	Canonical scalar kinetic-only has $w = +1$; signed-sector gradients can be gauge/representation artifacts	Correct stress tensor and separate physical scalar toy
Exact prediction of ϵ_Λ	Friedmann critical-density substitution uses observed H_0	Replace by dimensional and empirical calibration; no independent prediction
$V(\sigma_t) = \sigma_t^2/8 - \sigma_t^4/16$	No unique physical sector potential, dimensions, or microscopic normalization	Archive as trial shape, never governing QGD potential
Separate $\mathcal{Q}_{ab}$ and lowercase q_{ab} quantum sources	$\delta g^{\text{quantum}}$ is an on-shell response; historical $\mathcal{Q}$ still unproved	One active quantum-matter backreaction, no double count
$w(z)$ from arbitrary σ_t polynomial	Universal coupling and selected scalar degree not derived	State general continuity and source-vertex requirements
$\Delta\ell_1/\ell_1 = -f_\sigma/2$	Acoustic features depend on sound horizon, distance and perturbations	Remove claimed universal shift; require Boltzmann analysis
Radiation/FLRW solutions	Chapter 2 already establishes conformal Dirac radiation and a regular FLRW metric lift	Promote as central constructive benchmark without claiming microscopic emergence

21 Conclusions and forward decision

Established in this chapter. An exact fixed-flat-reference signed-sector representation exists for every smooth spatially flat FLRW scale factor. The Chapter 6 Misner–Sharp quasilocal-energy dictionary extends exactly to FLRW and obeys the comoving pressure-work law under the adopted effective Einstein dynamics. Chapter 5's unchanged static collective Poisson functional has no homogeneous order-parameter solution for a nonzero homogeneous source; a separate zero-mode or temporal-response model is required. An appropriately augmented regular branch spans metric variations and therefore pulls back the *adopted* Einstein–matter equations. The massless Dirac radiation benchmark is compatible with the same geometry and accepted Friedmann dynamics. The old kinetic-only $w = -1$ and direct observed-density "prediction" fail elementary stress-tensor and dimensional checks. A properly defined illustrative fourth-order scalar operator also differs from the displayed historical one. A time-dependent redundant signed-sector representative can carry nonzero naive derivative terms without changing the metric.

New v1.2 result. On the explicit collisionless, isotropic Chapter 1 free-Dirac kinetic branch, the homogeneous energy and pressure satisfy $\mathrm{d}\epsilon_D/\mathrm{d}\ln a = -3(\epsilon_D + p_D)$ and $0 \le p_D/\epsilon_D \le 1/3$. If the adopted flat Einstein equations with $\Lambda = 0$ have only this nonzero positive kinetic source, accelerated expansion is excluded. This is a conditional microphysical constraint, not a full quantum-vacuum no-go. The static $\boldsymbol{k} \neq 0$ susceptibility separately obeys the mean-zero Poisson boundary condition, but gives no $\boldsymbol{k} = 0$ equation. The remaining causal kernel must be computed with initial-state,

renormalization, Ward, spectral, and source-coupling tests.

New v1.3 result. The existing Chapter 1 scalar-contact Dirac model admits a regulated finite-temperature Hartree extension with a stationary gap $M(T)$ and exactly consistent homogeneous equilibrium thermodynamics. Adopting entropy conservation and the *separately supplied* flat Einstein equations gives a computable, explicitly conditional isentropic $a(t)$, which decelerates on the tested positive-pressure branch. In the stationary collisionless RPA homogeneous scalar channel, the positive bare mass and gap equation yield the exact bound $K_R(\omega, 0) \geq m_0/(2G_4M) > 0$ for $|\omega| \leq 2M$; no real subthreshold pole exists on this branch. The thermodynamic susceptibility and causal zero-frequency limit differ at finite temperature, so they must not be used interchangeably as a zero-mode evolution law. These conclusions constrain only the specified cutoff channel, not all collective matter modes.

New v1.4 result. The absolute flat-space Hartree vacuum density is not fixed by the microscopic saddle or thermal equation of state: $\Omega \mapsto \Omega + C$ is compensated in the effective gravitational action by a finite cosmological-constant shift. A separately rematched covariant proper-time Dirac determinant generates local $\sqrt{-g}$ and $\sqrt{-g}R$ coefficients, and a positive-curvature-Hessian Hartree saddle gives the explicit local condensate response $\delta M = -C'_R R/K$ and the conditional $-C'^2_R R^2/(2K)$ correction. These are regulator-level local results with unmatched finite gravitational counterterms, not physical predictions of G_N, Λ, or an inflationary R^2 scale. A separate exact $f(R) = R - 2\Lambda + \alpha R^2$ trace check shows $R = 4\Lambda$ for a constant-curvature vacuum; adding αR^2 does not itself select de Sitter when $\Lambda = 0$.

v1.6 stress-response result. At homogeneous zero momentum, an isolated stationary system has vanishing non-contact energy-density retarded commutator; the isotropic Hartree one-loop shear channel exhibits only a fermion-pair continuum above the threshold $2M$ and scalar–tensor mixing vanishes in the separable RPA. These are controlled exclusions of specific naive graviton identifications, not a proof that microscopic quantum matter cannot generate gravity.

Not established. Geometric energy reconstruction is not an independent selection law, and the static-bridge obstruction is not a no-go result for emergent cosmology. We have not derived a microscopic mechanism that chooses $a(t)$, a nonzero observable scalar potential, an inflationary trajectory, the late-time dark-energy scale, a definite $w(z)$, an autonomous healthy fourth-order σ equation, or any independent cosmological deviation from GR. The historical operator $\lambda\mathcal{Q}_{ab}$ remains part of the full founding metric, with no unique off-shell microscopic definition. Metric-only quantum backreaction remains encoded in the computed $\delta g^{\text{quantum}}$ response and is not added twice.

New chapter decision. The correct next step is not to fit the old σ_t source to a preselected expansion history. It is first to determine the homogeneous dynamical zero mode missing from the conditional Chapter 5 static model and then calculate a physically selected collective response of interacting quantum matter, including photons and finite-density fermions, whose low-energy cosmological stress, metric coupling and perturbations can be compared with observation. In this programme cosmology is both a useful exact representation benchmark and a potentially decisive test of QGD's microscopic matter-wave hypothesis.

22 A bottom-up energy-source vertex: return to Chapter 5

Research change. Chapter 5 already defines the overarching microscopic problem through $V_A{}^I[K^{-1}]_{IJ}V_B{}^J$. Rather than treating every cosmological response channel as a new fundamental emergence test, we now derive one *source-resolved* entry from the existing Chapter 1 fermion/contact functional. The primary proof is in Chapter 5 v3.4; the present section records its cosmological interpretation and the equilibrium boundary conditions.

Take the homogeneous zero-chemical-potential Hartree potential $\Omega(M, T)$ in Eq. (46), the thermally selected mass $M_*(T)$ in Eq. (47), and an *external static lapse* $ds^2 = -N^2dt^2 + d\mathbf{x}^2$. At fixed reference

temperature T_0 and fixed physical local momentum cutoff, equilibrium requires $T_{\rm loc} = T_0/N$. Define

$$\mathcal{F}(N,M) = N[\Omega(M, T_0/N) - \Omega_v], \qquad \Omega_M(M_*(T), T) = 0. \tag{73}$$

The exact homogeneous static mixed source vertex, obtained by *two derivatives of the same microscopic functional*, is

$$\boxed{\begin{aligned} V_{NM} &= \mathcal{F}_{NM}\big|_{N=1,M=M_*} = -T\Omega_{TM} \\ &= -\frac{2M_*}{\pi^2 T}\int_0^{\Lambda_c} dp\, p^2 n_p(1-n_p), \quad n_p = (e^{E_p/T}+1)^{-1}. \end{aligned}} \tag{74}$$

The relevant static collective Hessian is $K_{MM} = \Omega_{MM}$, *not* automatically the collisionless real-frequency kernel $K_R(\omega, \mathbf{0})$ tested in Section 17. At the calibrated cutoff benchmark $T = 0.4$ we find $M_* = 0.76699593$, $K_{MM} = 0.20651331$, and $V_{NM} = -0.02452384$, with finite exchange term $V_{NM}^2/K_{MM} = 0.00291225$ in the model's units.

The on-shell static response obeys the exact thermodynamic identity

$$\boxed{\begin{aligned} \mathcal{F}''_{\rm on}(1) &= \underbrace{T^2\Omega_{TT}}_{\text{direct/contact}} - \underbrace{\frac{V_{NM}^2}{K_{MM}}}_{\text{collective exchange}} = -Tc_V, \\ \Delta c_V^{\rm coll} &= \frac{V_{NM}^2}{TK_{MM}}, \qquad \left.\frac{dM_*}{dN}\right|_1 = -\frac{V_{NM}}{K_{MM}}. \end{aligned}} \tag{75}$$

This result verifies the static source–Hessian–vertex matching and reproduces the equilibrium heat-capacity correction. It does not derive a $1/|\mathbf{k}|^2$ pole, the observed cosmological constant, a self-consistent $H(t)$, or an autonomous QGD gravitational field. A spatially homogeneous static lapse source also depends on thermal boundary data; it cannot be equated with the retarded $k = 0$ energy commutator, which vanishes for a stationary isolated Hamiltonian apart from separately defined contact terms.

For the *physical* Chapter 7 cosmological response we must next (i) derive the finite-momentum stress–mass vertex, (ii) include the complete classical plus renormalized matter stress and its Ward contacts, (iii) account for the physical signed-sector quotient, and (iv) calculate retarded dynamics on a specified nonequilibrium FLRW state. This follows the Chapter 5 microscopic response programme rather than repeating it under a new cosmological name.

A Verification details and units

Algebraic signed lift. For $p = (d+b^2)/(2b)$ and $n = (d-b^2)/(2b)$, $p^2 - n^2 = d$. For spatially flat FLRW set $d = a^2 - 1$ in each spatial direction and independently add a cancelling time pair for regular perturbation theory. For $P = N = ve^{\theta} I_4$, $P^{\mathsf{T}}P - N^{\mathsf{T}}N = 0$ and $\|\dot{P}\|_F^2 + \|\dot{N}\|_F^2 = 8v^2 e^{2\theta}\dot{\theta}^2$. These identities have been checked with the bundled symbolic script.

Fourth-order identity. Differentiate $\mathscr{D}\varphi = \ddot{\varphi} + 3H\dot{\varphi}$ and apply $\mathscr{D}$ again. The result is Eq. (26); a covector calculation requires its own connections. For the illustrative constant-H trial, solve $r(r+3H)[1 - \ell^2 r(r+3H)] = 0$ to obtain Eq. (28).

Energy-density units. With H in s^{-1}, $[c^2H^2/G_N] = \mathrm{J\,m^{-3}}$. The historical use of c^4H^2/G_N is not an energy density in SI units. The pressure p and ϵ in Eqs. (10)–(11) both refer to energy per volume; no extra c^2 is inserted into the pressure term.

Source provenance. Source historical file: `Ch5-cosmology.tex`, titled "Chapter 5: Cosmology" (March 2026 historical narrative). Governing QGD corpus: Chapter 1 final QGD–075; Chapter 2 final QGD–075; Chapter 3 QGD–093; Chapter 4 QGD–141 notation-aligned edition; Chapter 5 v3.3 notation-aligned edition; Chapter 6 v1.8-F notation-aligned edition. References to historical formulas document an audit, not acceptance of their claims.

B Ward-constrained vacuum metric–mass source: a Chapter 5 result for cosmology

Chapter 5 v3.5 now derives the *leading long-wavelength, zero-temperature Euclidean* stress–collective-mass vertex in the separately rematched proper-time contact model. Its result is algebraically the metric variation of Chapter 7's earlier curvature coefficient and must not be interpreted as an extra operator. Let $M = M_v + \chi$ and $g_{\mu\nu} = \delta_{\mu\nu} + h_{\mu\nu}$. At the stationary Hartree vacuum,

$$\boxed{\begin{aligned} V^E_{\mu\nu,M}(k) &= C'_R(M_v)(k^2\delta_{\mu\nu} - k_\mu k_\nu) + O(k^4), \\ k_\mu V^E_{\mu\nu,M}(k) &= 0, \qquad C'_R(M_v) = 0.00452781351, \end{aligned}} \tag{76}$$

where the numerical value uses the *proper-time* benchmark $\Lambda_{\mathrm{PT}} = 4, m_0 = 0.12, M_v = 0.8$, not the three-momentum cutoff of the thermal calculation. Fermion-loop and Hartree-auxiliary k-independent contact terms cancel only after the complete stationary gap condition: $\frac{1}{2}U'_D = -0.13837051325$ and $\frac{1}{2}U'_{\mathrm{aux}} = +0.13837051325$. Therefore the static zero-temperature vacuum lapse component satisfies $V^E_{00,M}(0,\boldsymbol{k}) = C'_R(M_v)|\boldsymbol{k}|^2 + O(|\boldsymbol{k}|^4)$, but the distinct *finite-temperature* equilibrium lapse derivative V_{NM} of Sec. 22 need not vanish at $\boldsymbol{k} = 0$.

The scalar condensate has positive $K_v = 0.1479798341$, so its exchange contribution is $VK^{-1}V = O(k^4)$, and its transverse-traceless tensor projection vanishes at this order. Completing the square gives

$$\boxed{\Delta\Gamma^{(2)}_E = -\frac{[C'_R(M_v)]^2}{2K_v}\int \mathrm{d}^4x\,[\delta R_E]^2 + \cdots .} \tag{77}$$

This *recovers* Eq. (62), rather than adding a second R^2 term. It is no prediction of accelerated expansion: a dynamical gravitational sector, finite local counterterms, complete causal interacting stress and appropriate state/initial conditions are still needed. The operational gain is a microscopic *Ward-compatible* component of Chapter 5's $V_A{}^I$, now connected to Chapter 7's vacuum curvature calculation with no double counting.

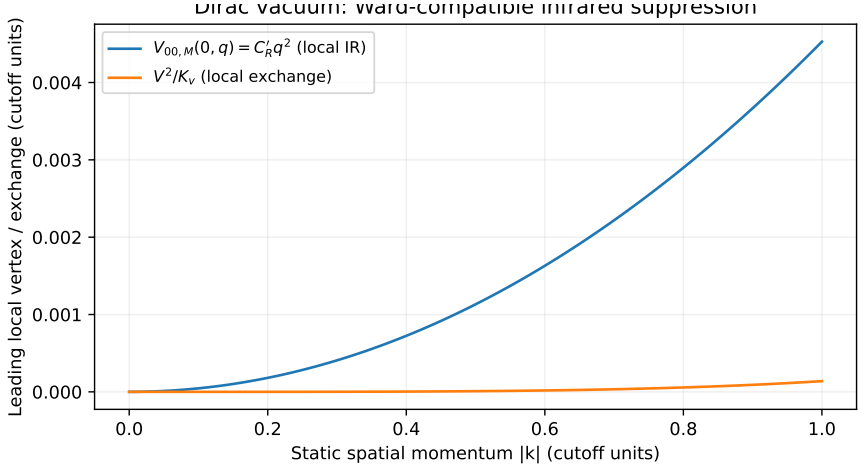


Figure 5: Vacuum local derivative-expansion vertex and exchange response: analytic infrared suppression rather than a candidate massless gravitational pole. Results outside the gradient-expansion regime are not claimed.

Next cosmological criterion. Derive the retarded $(\omega, \boldsymbol{k})$ stress/mass vertex with full contact and conserving identities on a specified stationary quantum state, then extend the physical source response to slowly evolving FLRW. A homogeneously expanding background is not a limit of a static Euclidean vacuum calculation taken without a specified state and in–in continuation. The

complete founding metric with $-\lambda\mathcal{Q}_{ab}$ is retained; current calculations fix $\lambda\mathcal{Q} = 0$ as a declared specialization.

C A finite-frequency microscopic vacuum vertex: implications for QGD cosmology

Chapter 5 v3.6 derives the subtracted full one-loop momentum dependence of the stationary *vacuum* mixed Dirac stress–Hartree-mass vertex. Unlike the earlier v1.7 result, which retained only $O(k^2)$, the new form factor resolves the two-fermion branch cut after analytic continuation:

$$\boxed{\operatorname{Im} F_R(\omega, \boldsymbol{k}) = \frac{M}{48\pi}\left(1 - \frac{4M^2}{\omega^2 - |\boldsymbol{k}|^2}\right)^{3/2} \Theta(\omega^2 - |\boldsymbol{k}|^2 - 4M^2) \quad (\omega > 0).} \tag{78}$$

At the matched reference vacuum mass $M_v = 0.8$, the zero-momentum pair threshold is $\omega_{\text{th}} = 1.6$ and the $|\boldsymbol{k}| = 0.6$ threshold is $\omega_{\text{th}} = 1.70880$ in cutoff-model energy units. The vertex tensor is identically transverse when all source contacts are included, and the covariant subtracted scalar form factor has no zero-momentum pole. Its small momentum behavior is

$$\boxed{V^E_{\mu\nu,M}(k) = \left(k^2\delta_{\mu\nu} - k_\mu k_\nu\right)\left[F_E(0) - \frac{k^2}{480\pi^2 M_v} + O(k^4)\right].} \tag{79}$$

With $K_v > 0$, the scalar exchange is $O(k^4)$ near the origin. This reproduces, rather than supplements, the v1.4 curvature-relaxation R^2 term at leading order; there is no additional independent gravitational sector. The finite-momentum cut determines the *possibility of vacuum pair absorption by a supplied metric perturbation*; it cannot select $H(t)$, Λ_{eff}, or the FLRW quantum state. The analytic continuation is valid for the zero-temperature stationary Minkowski vacuum and is *not* a calculation of the retarded stress response in a time-dependent FLRW state. **Status and next test.**

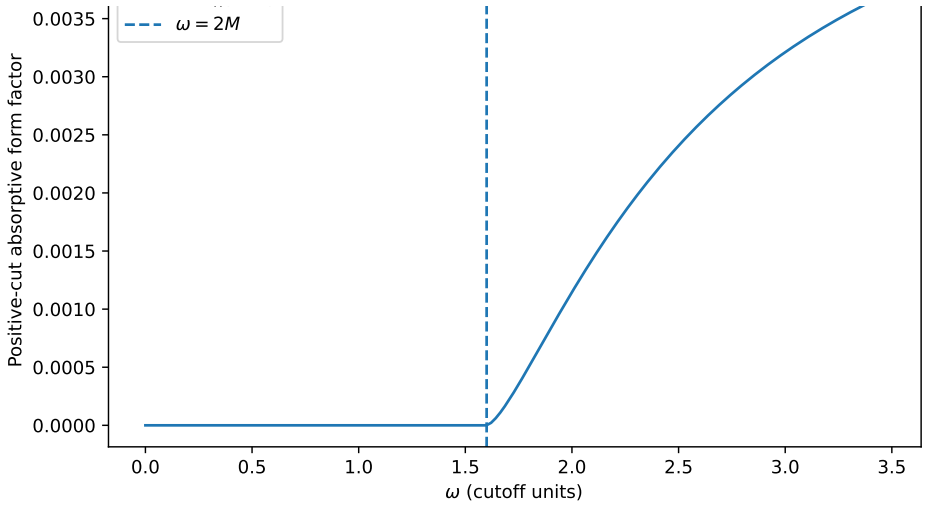


Figure 6: The computed massive-Dirac pair threshold. A real-time FLRW state requires a separate in–in calculation.

This is a [D] result for a well-specified one-loop, vacuum Dirac quasiparticle channel with separately matched local coefficient; all claims of independent QGD gravitational emergence remain [O]. Chapter 5's next calculation should include the fully momentum-dependent scalar collective kernel and, separately, a finite-temperature conserving source vertex. Neither the known thermodynamic v3.4 lapse coupling nor the zero-temperature vacuum result is a substitute for the required causal interacting FLRW stress tensor. No extra historical $\mathcal{Q}_{ab}$ has been inserted.

D Natural-invariant-first cosmology: a normalized quantum test (v1.9)

Terminology. We call contributions of $\hbar\Gamma^{(1)}_{\mathrm{m,ren}}$ to QGD's action and natural-invariant equations *QGD quantum corrections*; each term retains an explicit attribution to its calculated microscopic state (a conformally coupled massless fermion field for the example below). A different field species or state would give different coefficients. QGD terminology is not a claim that the anomaly is unknown to semiclassical GR. Unlike a guessed $\dot\sigma^2$ energy, every term below arises from the fixed QGD–136 action and its renormalized stress.

D.1 Normalized spatial and temporal signed-sector variables

For fixed Cartesian coframe and the controlled $\lambda\mathcal{Q}=0$ specialization, let $s_s=-\varepsilon_s$, and for each fixed spatial i without summation define

$$\boxed{\Xi_\sigma := \sum_s s_s(\sigma_i^{(s)})^2 = a^2-1, \qquad \Upsilon_\sigma := \sum_s s_s(\sigma_0^{(s)})^2 = 1-N^2.} \tag{80}$$

Homogeneous isotropy demands that the three spatial signed sums coincide. For the exact pair-lift already derived, both Ξ_σ and Υ_σ are independent of its arbitrary nonzero factorization constant. They are invariant under the signed transformations that leave the fixed-coframe metric intact, but are *not* standalone diffeomorphism-invariant observables. The reference-epoch convention $a(t_*)=1$ fixes $\Xi_\sigma(t_*)=0$; it does not fix a new gravitational coupling.

Define the proper-time normalized natural-invariant expansion and its square:

$$\boxed{\vartheta_\sigma = \frac{\dot\Xi_\sigma}{2\sqrt{1-\Upsilon_\sigma}(1+\Xi_\sigma)}, \qquad \mathcal{I}_\sigma := \vartheta_\sigma^2 = \frac{\dot\Xi_\sigma^2}{4(1-\Upsilon_\sigma)(1+\Xi_\sigma)^2}.} \tag{81}$$

Dots in this subsection refer to the coordinate t, and $d\tau = N dt = \sqrt{1-\Upsilon_\sigma}\,dt$. Standard FLRW notation would call ϑ_σ the Hubble rate, but the QGD natural-invariant coordinate is primary here.

D.2 Main action and Option C expressed in natural invariants

The full governing construction remains $g_{\mu\nu} = T^a{}_\mu T^b{}_\nu[\eta_{ab} - \sum_s \varepsilon_s \sigma_a^{(s)}\sigma_b^{(s)} - \lambda\mathcal{Q}_{ab}]$ and $\Gamma^{\mathrm{ren}}_{\mathrm{QGD}} = S_{\mathrm{EH}} + S_{\mathrm{GHY}} + S^{\mathrm{cl}}_{\mathrm{F1}} + \hbar\Gamma^{(1)}_{\mathrm{m,ren}} + O(\hbar^2)$. The full formal natural-invariant Euler equation remains

$$\boxed{2s_s\Pi^{ab}\sigma_b^{(s)} = \lambda[(D_s\mathcal{Q})^*_{\sqrt{-g}}\Pi]^a.} \tag{82}$$

In the controlled metric-only specialization, with fixed coframe and a finite comoving spatial reference volume, the gravitational minisuperspace Lagrangian (natural units) becomes

$$\boxed{L_{\mathrm{G}} = -\frac{3\dot\Xi_\sigma^2}{32\pi G_N\sqrt{1-\Upsilon_\sigma}\sqrt{1+\Xi_\sigma}} - \frac{\Lambda_{\mathrm{ren}}}{8\pi G_N}\sqrt{1-\Upsilon_\sigma}(1+\Xi_\sigma)^{3/2}.} \tag{83}$$

The matter sector is varied through the same metric, with its quantum state held and varied as required by its specified in–in prescription. Independent lapse and scale variations yield, after quantum-stress renormalization,

$$\boxed{3\mathcal{I}_\sigma = \Lambda_{\mathrm{ren}} + 8\pi G_N(\epsilon_{\mathrm{cl}} + \epsilon_{\mathrm{QGD,q}}), \quad 2\frac{d\vartheta_\sigma}{d\tau} + 3\mathcal{I}_\sigma = \Lambda_{\mathrm{ren}} - 8\pi G_N(p_{\mathrm{cl}} + p_{\mathrm{QGD,q}}).} \tag{84}$$

These equations agree with Einstein–Friedmann on the regular metric-only branch; Option C does not create an independent third homogeneous equation. The curvature scalar also takes the compact normalized form

$$\boxed{R = \frac{3}{(1+\Xi_\sigma)\sqrt{1-\Upsilon_\sigma}}\frac{d}{dt}\left(\frac{\dot{\Xi}_\sigma}{\sqrt{1-\Upsilon_\sigma}}\right).} \tag{85}$$

D.3 QGD quantum correction: one specified massless-fermion state

For one free conformal four-component Dirac field in the specified conformal state, with vanishing finite $\Box R$ trace term and no double counting of the same state in $S^{\rm cl}_{\rm F1}$, define $\beta = 11/(960\pi^2)$. Conservation and the established trace anomaly give

$$\boxed{\epsilon_{\rm QGD,q} = \beta\mathcal{I}_\sigma^2 + \frac{C_\varrho}{(1+\Xi_\sigma)^2}, \qquad p_{\rm QGD,q} = -\beta\mathcal{I}_\sigma^2 - \frac{4\beta}{3}\mathcal{I}_\sigma\frac{d\vartheta_\sigma}{d\tau} + \frac{C_\varrho}{3(1+\Xi_\sigma)^2}.} \tag{86}$$

Thus the normalized constraint is

$$\boxed{3\mathcal{I}_\sigma = \Lambda_{\rm ren} + 8\pi G_N\left[\epsilon_{\rm cl} + \frac{C_\varrho}{(1+\Xi_\sigma)^2}\right] + \gamma\mathcal{I}_\sigma^2, \qquad \gamma = \frac{11G_N}{120\pi}.} \tag{87}$$

There is no fitted sector potential, independent $\mathcal{Q}_{ab}$ or extra normalization. The state and $\Lambda_{\rm ren}$ remain independent inputs, and the correction is one specified contribution within standard semiclassical dynamics as organized by QGD.

D.4 A new normalized radiation-era relation

For a separately prescribed classical radiation source $\epsilon_{\rm cl} = \epsilon_*(1+\Xi_\sigma)^{-2}$, take $\Lambda_{\rm ren} = C_\varrho = 0$ and $N = 1$, and set $K = 8\pi G_N\epsilon_*/3$ and $\alpha = 11G_N/(360\pi)$. On the GR-connected perturbative branch of the constraint, with $\alpha K/(1+\Xi_\sigma)^2 \ll 1$,

$$\boxed{\frac{d\Xi_\sigma}{d\tau} = 2\sqrt{K}\left[1 + \frac{\alpha K}{2(1+\Xi_\sigma)^2}\right] + O(\alpha^2).} \tag{88}$$

Choosing $\Xi_\sigma(\tau_*) = 0$ and integrating yields

$$\boxed{\tau - \tau_* = \frac{\Xi_\sigma}{2\sqrt{K}} - \frac{\alpha\sqrt{K}}{4}\frac{\Xi_\sigma}{1+\Xi_\sigma} + O(\alpha^2).} \tag{89}$$

At zero quantum correction $\Xi_\sigma = 2\sqrt{K}(\tau - \tau_*)$ is *linear* in proper time, while the original scale factor is its square root. The first quantum term bends this linear natural-invariant law. This is a calculated, normalized QGD *representation* of the standard semiclassical correction, not a new independent quantum gravitational effect. Its extrapolation to high curvature is outside one-loop control.

Energy and stopping rule. On a comoving sphere of areal radius $R_{\rm areal} = r_c\sqrt{1+\Xi_\sigma}$, the metric-only Misner–Sharp energy is $E_{\rm MS} = R^3_{\rm areal}\mathcal{I}_\sigma/(2G_N)$. The corresponding source balance follows from Eq. (87). Purely metric-preserving changes of sector representatives change neither Ξ_σ nor this energy. An additional QGD-specific cosmological prediction would require a derived sector-sensitive functional, physically normalized state selection, or other independently determined dynamics; it cannot be extracted from an arbitrary signed representative amplitude.

E Final normalization, energy closure, and the independent-prediction criterion (v2.0)

Purpose and no-new-term rule. This section consolidates already-established QGD structures rather than postulating a new sector action, cosmological potential, or source field. The governing action is exactly Eq. (54), the complete founding metric is Eq. (2), and the preferred natural-invariant field equation is Eq. (3). The renormalized QGD quantum source is varied once through the same action. The controlled FLRW calculation remains the fixed-coframe, $\lambda\mathcal{Q} = 0$ metric-only specialization; this does not assert that the unmatched historical $\mathcal{Q}_{ab}$ vanishes fundamentally.

E.1 Relative epoch normalization and physical observables

In the FLRW chart of Section D, $\Xi_\sigma = a^2 - 1$ and $\Upsilon_\sigma = 1 - N^2$ are signed-sector combinations, independent of the signed-factorization parameter at fixed reference coframe. Neither an individual $\sigma_\mu^{(s)}$ nor its coordinate derivative is intrinsically observable. Fix a reference proper-time epoch τ_* for one comoving congruence and define

$$\boxed{Y_\sigma(\tau) := \frac{1+\Xi_\sigma(\tau)}{1+\Xi_\sigma(\tau_*)}, \quad \mathcal{A}_\sigma = \sqrt{Y_\sigma}, \quad \mathcal{N}_\sigma = \tfrac{1}{2}\ln Y_\sigma, \quad \vartheta_\sigma = \frac{d\mathcal{N}_\sigma}{d\tau}.} \tag{90}$$

Here $d\tau = \sqrt{1-\Upsilon_\sigma}\,dt$ and $Y_\sigma = (a/a_*)^2$. Thus for comoving emitter and observer with unchanged local-clock conventions,

$$\boxed{1 + z_{e\to 0} = \sqrt{\frac{1+\Xi_\sigma(\tau_0)}{1+\Xi_\sigma(\tau_e)}} = \sqrt{\frac{Y_\sigma(\tau_0)}{Y_\sigma(\tau_e)}}.} \tag{91}$$

The ratios above are independent of a constant rescaling of comoving spatial coordinates when the reference coframe is transformed consistently. They are observable through specified propagation and clock measurements, not because the individual natural-invariant representatives are new degrees of freedom.

Proposition E.1 (Exact constant-equation-of-state energy closure [D] | [C]). *Assume a separately conserved homogeneous perfect-fluid component with $p_w = w\epsilon_w$, constant w, and the usual FLRW comoving volume. Then*

$$\boxed{\frac{\epsilon_w}{\epsilon_{w,*}} = Y_\sigma^{-3(1+w)/2}, \qquad \frac{E_w}{E_{w,*}} = Y_\sigma^{-3w/2}, \qquad dE_w = -p_w\,dV.} \tag{92}$$

Proof. The covariant continuity equation gives $d\epsilon_w/d\mathcal{N}_\sigma = -3(\epsilon_w + p_w)$ and $\mathcal{N}_\sigma = \ln(a/a_*)$. Integrating with constant w yields $\epsilon_w/\epsilon_{w,*} = \mathcal{A}_\sigma^{-3(1+w)}$. The physical volume of a fixed comoving region obeys $V/V_* = \mathcal{A}_\sigma^3$; therefore $E_w/E_{w,*} = \mathcal{A}_\sigma^{-3w}$. Differentiating $E = \epsilon V$ and applying continuity gives the work law. □

Source	$\epsilon_w/\epsilon_{w,*}$	$E_w/E_{w,*}$
Dust ($w = 0$)	$Y_\sigma^{-3/2}$	1
Radiation ($w = 1/3$)	Y_σ^{-2}	$Y_\sigma^{-1/2}$
Specified vacuum fluid ($w = -1$)	1	$Y_\sigma^{3/2}$

The above energy ratios are standard covariant conservation expressed in normalized QGD variables, not independently derived new forces. The homogeneous Misner–Sharp energy of a sphere with comoving radius r_c, physical areal radius $R_{\rm ar} = r_c\sqrt{1+\Xi_\sigma}$, and zero spatial curvature is

$$\boxed{E_{\rm MS}(r_c) = \frac{R_{\rm ar}^3\mathcal{I}_\sigma}{2G_N} = \frac{r_c^3}{8G_N\sqrt{1+\Xi_\sigma}}\left(\frac{d\Xi_\sigma}{d\tau}\right)^2.} \tag{93}$$

This geometric energy includes the effects of every source entering the constraint, including $\Lambda_{\rm ren}$ if present. Its meaning must not be conflated with Chapter 6's static, reference-space scalar energy $E[H;\rho_m]$, nor may a cosmological constant be counted both geometrically and as a duplicate matter source.

E.2 One-epoch calibration and the controlled QGD quantum branch

For a single separately conserved fluid $w > -1$ with $\Lambda_{\rm ren} = C_\varrho = 0$, take the specified one-conformal-massless-fermion anomaly contribution in Eq. (86), including its stated state and renormalization assumptions. Define

$$\boxed{n := \frac{3(1+w)}{2}, \quad K := \frac{8\pi G_N \epsilon_{w,*}}{3}, \quad \alpha_Q := \frac{11 G_N}{360\pi}, \quad \mathcal{I}_{\sigma,*} := \vartheta_\sigma^2(\tau_*).} \tag{94}$$

The perturbatively GR-connected solution of Eq. (87) is

$$\mathcal{I}_\sigma(Y_\sigma) = K Y_\sigma^{-n} + \alpha_Q K^2 Y_\sigma^{-2n} + O(\alpha_Q^2). \tag{95}$$

Use the expansion normalization at the reference epoch once: $\mathcal{I}_{\sigma,*} = K + \alpha_Q K^2 + O(\alpha_Q^2)$. Eliminating K gives the dimensionless cross-epoch relation

$$\boxed{\frac{\mathcal{I}_\sigma(Y_\sigma)}{\mathcal{I}_{\sigma,*}} = Y_\sigma^{-n}\left[1 + \alpha_Q \mathcal{I}_{\sigma,*}\big(Y_\sigma^{-n} - 1\big)\right] + O(\alpha_Q^2).} \tag{96}$$

The approximation assumes $|\alpha_Q \mathcal{I}_\sigma| \ll 1$ throughout the interval, a controlled derivative expansion, and no additional renormalized curvature/state terms omitted at the same order. Here "QGD quantum correction" denotes the specified matter quantum contribution *within the QGD action*; its underlying anomaly is also part of established semiclassical gravity and is not claimed as an exclusive prediction of QGD.

The proper-time acceleration follows without a new sector force law:

$$\boxed{\frac{1}{a}\frac{d^2 a}{d\tau^2} = (1-n) K Y_\sigma^{-n} + (1-2n)\alpha_Q K^2 Y_\sigma^{-2n} + O(\alpha_Q^2).} \tag{97}$$

Consequently the controlled benchmark still decelerates for isolated dust ($n = 3/2$) and radiation ($n = 2$). A separately specified positive vacuum source may accelerate, but this procedure neither fixes its magnitude nor explains its microscopic origin. The formal Planck-curvature root of the truncated constraint lies outside the controlled one-loop regime.

E.3 No extra homogeneous field equation on regular metric-only branches

Theorem E.2 (Conditional equivalence of the metric and natural-invariant Euler systems [C])**.** *Let* $\Gamma^{\rm ren}_{\rm QGD}[\sigma, \text{matter}] = \Gamma^{\rm ren}_{\rm metric}[g(\sigma), \text{matter}]$ *with no additional independent sector functional. Suppose the full differentiable metric map has a right inverse at the background on the admissible variation spaces, including prescribed boundary conditions. Then*

$$\boxed{(D_\sigma g)^* \frac{\delta \Gamma^{\rm ren}_{\rm metric}}{\delta g} = 0 \quad \Longleftrightarrow \quad \frac{\delta \Gamma^{\rm ren}_{\rm metric}}{\delta g} = 0.} \tag{98}$$

The result applies to the entire renormalized action, including its QGD quantum stress; for a nonzero derivative-dependent $\mathcal{Q}_{ab}$ *it requires the right-inverse condition for the* full *differential map. In regular FLRW reductions spanning independent lapse and spatial scale-factor variations, Option C provides the two metric equations and no third independent homogeneous evolution equation.*

Proof. Write $J = D_\sigma g$ and $E_g = \delta\Gamma/\delta g$, with the functional adjoint defined under the same measure and boundary conditions. The chain rule gives $E_\sigma = J^* E_g$. If $JR = I$ for an admissible right inverse R, then $R^* J^* = I$ and hence $J^* E_g = 0$ implies $E_g = 0$. The reverse implication is immediate. The claim is conditional on the specified functional spaces and does not imply that a microscopic quantum action has independently generated the gravitational metric action. □

For the FLRW minisuperspace lapse N and scale factor a, the time-reparametrization Noether identity of the gravitational action, using its unnormalized Euler derivatives $\mathcal{E}_N := \delta\Gamma/\delta N$ and $\mathcal{E}_a := \delta\Gamma/\delta a$, is

$$\boxed{\dot{a}\,\mathcal{E}_a - N\frac{d\mathcal{E}_N}{dt} = 0.} \tag{99}$$

For the complete renormalized action, matter Euler derivatives and state-consistent conservation terms enter the corresponding identity. This is the constraint redundancy behind the absence of a separate new acceleration law on the regular metric-only branch.

Remark E.3 (A singular branch is not automatically new physics). In the fixed-coframe quadratic $\lambda\mathcal{Q} = 0$ specialization, $\delta g_{\mu\nu} = 2\sum_s s_s \sigma^{(s)}_{(\mu}\delta\sigma^{(s)}_{\nu)}$. At the all-zero sector background $\sigma^{(s)} = 0$, the linear map vanishes: $D_\sigma g = 0$. Hence sector stationarity is automatic there even when $\delta\Gamma/\delta g \neq 0$. This is a rank-deficient field redefinition, not by itself a solution of the metric equations or evidence for a new cosmological phase. Any exceptional branch requires independent physical and constraint analysis.

E.4 Representative redundancy and the final test of new physics

For a fixed spatial signed pair with $\Xi_\sigma = 3$ (so $a = 2$), the exact pair lift of Proposition 3.1 may use $\omega = 1$, giving $(\sigma_+, \sigma_-) = (2, 1)$, or $\omega = 2$, giving $(7/4, -1/4)$. Both have $\sigma_+^2 - \sigma_-^2 = 3$. Nevertheless for fixed ω and a varying $a(t)$,

$$\boxed{\dot{\sigma}_+^{\,2} + \dot{\sigma}_-^{\,2} = \frac{2a^2\dot{a}^2}{\omega^2}.} \tag{100}$$

Thus a naive "sector kinetic energy" built from this sum changes when an unobservable factorization parameter changes. No additional physical energy may be assigned to it without a derived sector-sensitive action and a justified observable or quotient. This is compatible with Chapter 6's stronger pure-orbit gradient-energy counterexample.

Final prediction boundary. Compare QGD and semiclassical GR using identical observable matter inputs, quantum state, renormalization prescription, and boundary data. On corresponding regular metric-only branches, their expansion and all metric-based observables coincide. Wherever $\mathcal{I}_{\mathrm{scGR}} > 0$, this is expressed by

$$\boxed{\mathcal{D}(z) := \frac{\mathcal{I}_{\sigma,\mathrm{QGD}}(z)}{\mathcal{I}_{\mathrm{scGR}}(z)} - 1 = 0.} \tag{101}$$

A future nonzero $\mathcal{D}$ would require a demonstrably additional physical restriction, source coupling, selection law, or independent sector-sensitive functional; merely relabeling a known semiclassical response cannot produce it. A distinct prediction must also be invariant under the physically redundant signed-sector transformations, preserve conservation and stability, and recover GR in its specified limit. These requirements are the stopping rule for the present research edition. No extra dynamical operator or cosmological parameter is introduced here.

F Q053–Q054: The quantum cosmological test and whether QGD dynamics produces dark energy (5 November 2026)

Aim. Decide what the present two-axiom, action-defined QGD model actually explains about late acceleration; then test the 2026 historical `Ch5-cosmology(1).tex` as an idea source. This update uses the QN004–QN005 action, Chapter 7's quantum-energy constraints, the full master metric (2), the v2.0 signed-sector cosmology and current observational results. It neither inserts nor silently restores the historical fourth-order field equation.

F.1 The three logically distinct meanings of "QGD dark energy"

There are three claims that must not be conflated.

1. **Existence of accelerated QGD solutions [D/C].** The controlled action contains $\Lambda_{\rm ren}$. For $\Lambda_{\rm ren} > 0$ a regular FLRW natural-invariant lift of de Sitter or ΛCDM exists. In natural units, $3H^2 = 8\pi G_N \rho_{\rm other} + \Lambda_{\rm ren} + \cdots$ and $\rho_\Lambda = \Lambda_{\rm ren}/(8\pi G_N)$ with $p_\Lambda = -\rho_\Lambda$. This is a legitimate QGD cosmological solution but not a derivation of the magnitude of $\Lambda_{\rm ren}$.

2. **Quantum-state vacuum response [C].** The renormalized $\Gamma^{(1)}_{\rm matter}$ supplies conserved $\langle T_{\mu\nu}\rangle_{\rm ren}$, state- and counterterm-dependent zero modes, and the controlled conformal anomaly. This can affect FLRW expansion but does not automatically yield a slowly evolving late-time component.

3. **Autonomous natural-invariant dark-energy dynamics [A/O].** A kinetic term, potential, or constitutive condition that distinguishes the signed-sector orbit would create additional dynamics. None has been established by the currently adopted metric-only action. It would need an independent variational specification and observable/representation treatment.

F.2 A conditional local-action no-self-acceleration theorem

Proposition F.1 (No new constant-curvature de Sitter branch from the stated local quadratic terms)**.** *Assume four-dimensional flat-reference regular QGD, $\lambda\mathcal{Q} = 0$, a complete metric-map differential, $\Lambda_{\rm ren} = 0$, vanishing classical and state-dependent stress, constant local couplings, standard bulk boundary conditions, and a low-curvature gravitational action consisting of Einstein–Hilbert plus $\alpha_R R^2 + \alpha_C C^2 + \alpha_E E_4$. Then no nonzero constant-curvature FLRW de Sitter solution follows from the local bulk Euler equations.*

Proof. A four-dimensional de Sitter metric obeys $R_{\mu\nu} = (R/4)g_{\mu\nu}$ with $R = 12H^2$ constant and $C_{\mu\nu\rho\sigma} = 0$. The R^2 Euler tensor

$$\mathsf{H}^{R^2}_{\mu\nu} = 2RR_{\mu\nu} - \tfrac{1}{2}g_{\mu\nu}R^2 + 2(g_{\mu\nu}\Box - \nabla_\mu\nabla_\nu)R \tag{102}$$

vanishes because $2R(R/4) - R^2/2 = 0$. The bulk variation of C^2 vanishes on an Einstein metric (Bach tensor), and E_4 is topological in four dimensions for a constant coupling. Thus the remaining vacuum equation is $G_{\mu\nu} = 0$, which a nonzero H de Sitter metric does not satisfy. Equivalently, for $f(R) = R + \beta R^2$, the constant-curvature trace $Rf_R - 2f = -R = 0$ requires $R = 0$. Regular metric-map completeness pulls this statement back to the natural-invariant field equations. □

This is a *conditional theorem*, not a no-go for nonlocal quantum states, higher-than-quadratic operators, degenerate sectors, or a specified independent sector-sensitive action. Analytic $R\mathcal{F}(\Box)R$ terms at constant R reduce to a local R^2 contribution if $\mathcal{F}(0)$ is regular and the nonlocal state/boundary prescription adds no further contribution; genuinely nonanalytic infrared response requires a separate test.

F.3 The one-loop Dirac anomaly: a formal Planckian branch, not today's dark energy

In the state and convention of the v1.9 radiation analysis, the homogeneous constraint contains

$$\boxed{3H^2 = 8\pi G_N \rho_{\rm cl} + \Lambda_{\rm ren} + \frac{11 G_N}{120\pi} H^4, \qquad c = \hbar = 1.} \tag{103}$$

For $\rho_{\rm cl} = \Lambda_{\rm ren} = 0$, the truncated algebraic equation has

$$\boxed{H = 0 \quad \text{or} \quad H^2 = \frac{360\pi}{11 G_N}.} \tag{104}$$

On the second branch $H t_P = \sqrt{360\pi/11} \simeq 10.14$, where the derivative and loop expansion that produced the term is uncontrolled. The formal root is therefore not an accepted physical inflationary or late-time dark-energy solution. At the representative observational $H_0 = 68.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$,

$$\boxed{\frac{\Delta H^2}{H^2} \simeq \frac{11}{360\pi}(H_0 t_P)^2 \simeq 1.39 \times 10^{-124}.} \tag{105}$$

No observable late-time $w(z)$ follows from this term. Its early-universe radiation-era effect is already given by the calculated Ξ_σ response of Section D. The Kerr quantum trace from Chapter 7 is a separate rotating background check, not a candidate homogeneous vacuum density.

F.4 The Hartree vacuum and the cosmological zero mode

The Chapter 5–6 regulated Dirac/Hartree determinant includes a homogeneous vacuum energy, but the renormalized constant part is shiftable against the local $\sqrt{-g}$ counterterm. In particular, a positive stationary Hartree mass, positive static scalar Hessian, or a gap in the scalar channel does not determine $\rho_{\rm DE,0}$. The full quantum action can be written schematically $\rho_{\rm vac}^{\rm phys} = \rho_{\rm bare} + \rho_{\rm loop} + \rho_{\rm finite\,ct}$; only the chosen renormalized sum gravitates. The QN004 curvature form factors start at higher curvature/gradient order once local terms are matched. They do not fix the Λ coefficient or turn a metric-preserving sector orbit into a fluid.

F.5 Legacy paper re-audit: what survives and what fails

The supplied historical *Chapter 5: Cosmology* suggests autonomous homogeneous natural-invariant dynamics, a fourth-order mode, a pure kinetic $w = -1$ attractor and a quartic potential $V_{\rm old}(s) = s^2/8 - s^4/16$. These are useful *questions*, not verified action consequences.

1. A *single temporal* signed-sector covector on a fixed flat coframe changes the lapse, not all spatial metric components: it cannot by itself produce the general expanding FLRW scale factor. The current Chapter 2 and v2.0 signed spatial lift correct this.

2. For a homogeneous standard positive-kinetic scalar with $K = \dot\phi^2/2$ and no potential, $p = \rho = K$ and $w = +1$, not $w = -1$. A nonzero constant $\dot\phi$ in de Sitter also violates $\ddot\phi + 3H\dot\phi = 0$ unless additional source dynamics is specified. The old sign assignment is not derived from a covariant stress tensor.

3. With constant H, the legacy fourth-order differential polynomial differs from its claimed factorization by $-3H^2\ell_Q^2\partial_t$ (with ℓ_Q understood as the appropriate time scale). The covariant scalar formula for $\Box_g^2\phi$ is already corrected in Section 9; a covector requires additional connection terms.

4. The old density "match" inserted the measured H_0 into $3H_0^2/(8\pi G_N)$: that is a critical-density identity, not a prediction of the dark-energy density or Ω_Λ. A separate source, normalization and energy-accounting calculation would be needed.

5. The polynomial $V_{\rm old}$ is dimensionless and depends on a chosen sector amplitude. At $s = \pm 1$ it has a local *maximum*: $V_{\rm old}(\pm 1) = 1/16$, $V'_{\rm old}(\pm 1) = 0$, $V''_{\rm old}(\pm 1) = -1/2$. It does not fix an energy scale or define a representation-invariant field potential. Its far-field static origin does not fix its cosmological action.

F.6 Two explicitly stated additional-action tests, neither a QGD prediction

Trial A: potential-free kinetic condensate. Suppose a *new physical, integrable* clock-type natural-invariant mode is identified, with a dimensionful scalar ϕ, $X = -\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi > 0$, and independently specified direct action $\Delta\Gamma_A = \int \sqrt{-g}\, P(X)$,

$$\boxed{P(X) = -X + \frac{X^2}{M_*^4}, \qquad \rho = 2XP_X - P, \quad p = P.} \tag{106}$$

At $X_* = M_*^4/2$ one has $P_X = 0$, $\rho_* = M_*^4/4 > 0$ and $p_* = -\rho_*$. The shift-current equation $\nabla_\mu (P_X \nabla^\mu \phi) = 0$ admits a constant-X_* homogeneous branch. This is a constructive *existence demonstration* of a pure-kinetic $w = -1$ solution in a separately chosen scalar effective theory, not a derivation from the current QGD axioms. Perturbations at $P_X = 0$ have vanishing leading sound-speed squared and require healthy higher-spatial-gradient completion; identifying ϕ with a physical QGD invariant rather than a redundant sector label is essential. Matching $\rho_* \simeq 5.3 \times 10^{-10}\,\mathrm{J\,m^{-3}}$ requires $M_* \simeq 3.18\,\mathrm{meV}$, a newly chosen scale rather than a prediction.

Furthermore, the homogeneous first integral is $a^3 P_X \dot{\phi} = C$. Near X_*, $X - X_* \propto a^{-3}$, hence on the healthy-side, $w + 1 \propto a^{-3} > 0$. A local CPL linearization about $a = 1$ then has $w_a \simeq 3(1 + w_0) > 0$, *opposite* to the $w_0 > -1, w_a < 0$ trend in some 2025–26 DESI combinations. A new interaction could change that behavior, but the bare trial should not be advertised as a DESI explanation.

Trial B: properly normalized legacy hilltop. If an independently identified physical scalar degree of freedom s exists, the old *shape* could be tested with

$$\boxed{\Delta\Gamma_B = -\int d^4x \sqrt{-g} \left[\frac{f^2}{2} g^{\mu\nu} \partial_\mu s \partial_\nu s + U_* \left(\frac{s^2}{8} - \frac{s^4}{16} \right) \right].} \tag{107}$$

The hilltop near $s = 1$ has $\rho \simeq U_*/16$, $m_s^2 = -U_*/(2f^2)$, and, for a conventional positive kinetic term, $w \geq -1$. Setting $U_* \simeq 16\rho_{\rm DE,0}$ and $|m_s| \sim H_0$ would require $f \sim \sqrt{24\Omega_{\rm DE,0}}\, M_{\rm Pl,red} \sim 4 M_{\rm Pl,red}$. A near-rest hilltop can *qualitatively* realize late-time thawing ($w_0 > -1, w_a < 0$), but the shape, field identity, U_*, f, initial conditions and perturbation stability are not predicted by QGD. The potential is unbounded below at large $|s|$ unless a completion is supplied. This is a transparent phenomenological model test, *not* the original paper's no-free-parameter result.

F.7 Observational gate: latest available results at 5 November 2026

DESI DR2 BAO and some combined supernova/CMB analyses continue to show fit-level hints of evolving dark energy; a 2026 combined analysis reports about 3σ in selected $w_0 w_a$ fits but no decisive Bayesian evidence (https://doi.org/10.1093/mnras/stag1607). The July 2026 DESI DR2 Lyα full-shape measurement shifts its central constraint toward ΛCDM (https://www.desi.lbl.gov/2026/07/30/new-desi-dr2-lyman-alpha-results-shed-light-on-dark-energy/). Dark Energy Survey Year 6 clustering/weak-lensing work accepted 2 November 2026 finds $w = -0.981^{+0.021}_{-0.022}$ in a joint constant-w analysis, with no significant preference over $w = -1$ (https://doi.org/10.1103/nnlj-gyr2). These datasets are not a detection of any particular natural-invariant field. The action-first QGD baseline ΛCDM branch remains compatible; its fixed H^4 correction is observationally negligible. Trial B can be fitted only after supplying a complete distinct action and perturbation model; no QGD-specific observational fit is claimed.

F.8 Conclusion and an actionable decision criterion

Established: The present QGD action admits dark energy through $\Lambda_{\rm ren}$, and quantum matter modifies cosmological natural-invariant solutions. **Not established:** a dynamical late-time $w(z)$ or a derived $\rho_{\rm DE}$ from its currently calculated curvature/Hartree terms alone. **Conditional existence:** an additional properly normalized, physically distinguished scalar natural-invariant mode can in principle drive de Sitter or thawing expansion; such an extension is ordinary scalar-tensor/effective-fluid physics until QGD's action and field-content selection provide further constraints. The first decisive test is not a new fit but a *fully specified covariant field-space observable*, direct sector action, constraint/perturbation stability analysis and normalization consistent with the classical QGD/GR branch. Do not count Λ, a Hartree vacuum constant and a selected sector potential as separate physical energy sources unless the action proves they are independent.

G Q055–Q058: Physical-sector selection and dynamical dark energy

Status and provenance (5 November 2026). This section integrates the stand-alone Q055–Q057 investigations and new Q058 analytic/numerical tests. It does not change either founding axiom, the complete master metric in Eq. (2), or the established renormalized metric-only action. The extensions below are tested *additional* actions, not terms derived from QGD's present axioms. We use $c = \hbar = 1$ and the reduced Planck mass $M_{\rm Pl}^2 = (8\pi G_N)^{-1}$ unless indicated. The quantum Dirac anomaly in Section F remains distinct and is never added twice.

G.1 Q055: An independent sector amplitude can be defined, but is not selected

In the declared fixed-reference specialization $\lambda\mathcal{Q} = 0$, let Y collect the signed natural invariants and $C = \mathrm{diag}(s_s)$. The metric deformation is $h = Y^{\mathsf{T}}CY$. A null signed pair has $Y_+ = Y_- = v\varphi(x)u$, hence $Y_+^{\mathsf{T}}Y_+ - Y_-^{\mathsf{T}}Y_- = 0$ even for varying φ. Q055 constructed a background with eight regular signed sectors supplying ten independent metric variations plus one additional independent amplitude direction: $\mathrm{rank}\, D(g,\varphi) = 11$. This is a *kinematic* independence result, not proof of a dynamically physical modulus. Q055's bounded, positive-kinetic hilltop test used

$$\Delta\Gamma_\varphi = -\int \mathrm{d}^4x\sqrt{-g}\left[\frac{f^2}{2}(\nabla\varphi)^2 + \frac{U_*}{16}\,\mathrm{sech}^2(\varphi^2-1)\right]. \tag{108}$$

With the particular FLRW initial conditions and normalization used there, numerical examples gave $(w_0, w_a) \simeq (-0.953, -0.087)$ and ordinary smooth-quintessence growth suppression at the sub-percent level. Q056 checked positive canonical high-frequency kinetic energy, $c_s^2 = 1$, and a model-dependent suppression of the present growth amplitude D by $\sim 0.62\%$ and of fD by $\sim 0.74\%$ for that representative case (equal early initial normalization). None of those numbers is a data fit or an independent QGD prediction.

G.2 Q056–Q057: The exact metric-only selection obstruction

Theorem G.1 (Null-pair nonselection in metric-mediated QGD [D]). *On a regular $\lambda\mathcal{Q} = 0$ branch, suppose the full action, including quantum matter, depends on Y only through $g(Y)$, and matter couples universally to this common metric. For a cancelling pair $Y_+ = Y_- = v\varphi u$, with $\varphi > 0$, neither the signed-sector metric, metric-coupled matter stress, its metric-mediated quantum corrections, nor fixed boundary values determine a unique interior φ.*

Proof. For $C_2 = \mathrm{diag}(1,-1)$, the boost $O(\chi) = \left(\begin{smallmatrix}\cosh\chi & \sinh\chi\\ \sinh\chi & \cosh\chi\end{smallmatrix}\right)$ obeys $O^{\mathsf{T}}C_2O = C_2$ and sends (φ,φ) to $(e^\chi\varphi, e^\chi\varphi)$. Even a spatially varying $\chi(x)$ leaves the metric and every metric-mediated action exactly unchanged. Moreover $D_Y g\,\delta Y = 0$ implies $D_Y\Gamma\,\delta Y = 0$ by the chain rule. Choosing smooth

χ of compact support preserves all boundary values while changing the interior amplitude. Thus no potential depending nontrivially on the pair amplitude is selected by these data. □

Q057 also found that the naive signed kinetic trial $-(f^2/2)[(\partial q_+)^2 - (\partial q_-)^2]$ vanishes identically on $q_+ = q_-$; as a theory of two unconstrained independent scalar amplitudes, its kinetic matrix has one wrong-sign eigenvalue. Hence the founding sign matrix cannot by itself supply a healthy positive kinetic metric. These are results about the *defined metric-only theory*, not a no-go for all conceivable direct sector interactions or derivative $\mathcal{Q}_{ab}$ models.

G.3 Q058: Least additional label-neutral sector structure

A controlled extension can be written without picking a + versus − field as the dark-energy variable, but *not* without adding physical structure. For one chosen $O(1,1)$ internal block define an independent positive-definite generalized sector metric

$$\mathsf{H}(\theta) = \begin{pmatrix} \cosh(2\theta) & \sinh(2\theta) \\ \sinh(2\theta) & \cosh(2\theta) \end{pmatrix}, \quad \mathsf{H}C_2\mathsf{H} = C_2, \quad \mathsf{H} > 0. \tag{109}$$

Its compatible scalar kinetic invariant is

$$\boxed{X_{\mathsf{H}} := -\frac{f^2}{16} g^{\mu\nu} \operatorname{tr}\big(\mathsf{H}^{-1}\partial_\mu \mathsf{H}\, \mathsf{H}^{-1}\partial_\nu \mathsf{H}\big) = -\frac{f^2}{2} g^{\mu\nu}\partial_\mu\theta\partial_\nu\theta.} \tag{110}$$

Global internal boosts act by $\theta \mapsto \theta + \chi$; a function of X_{H} preserves this shift and uses neither individual signed-sector label. A *local* boost $\chi(x)$ does not leave the derivative term invariant without a sector connection. Consequently declaring H physical reduces the former local metric-representation freedom to a specified global symmetry. The choice of the block, the new H, f, and the direct action are [A], not consequences of Axiom I or II. This is a conventional additional physical mode even though its coordinates are naturally adapted to QGD signed sectors.

G.4 Q058 trial A: shift-symmetric condensate and an early-radiation exclusion test

Consider the extra *global-shift-invariant* action

$$\boxed{\Delta\Gamma_{\mathsf{H}}^{(A)} = \int \mathrm{d}^4x\sqrt{-g}\, P(X_{\mathsf{H}}), \qquad P(X) = -X + \frac{X^2}{M_*^4}.} \tag{111}$$

With $y := X/M_*^4$ the exact homogeneous energy density, pressure, equation of state and adiabatic fluctuation speed are

$$\boxed{\rho_\theta = M_*^4(-y + 3y^2), \quad p_\theta = M_*^4(-y + y^2), \quad w = \frac{y-1}{3y-1}, \quad c_s^2 = \frac{2y-1}{6y-1}.} \tag{112}$$

For $y > 1/2$ one has positive $\rho_X = 6y - 1$, positive $P_X = 2y - 1$, $c_s^2 > 0$ and $w > -1$. At the exact condensate $y_* = 1/2$, $\rho = M_*^4/4$, $p = -\rho$, while $P_X = c_s^2 = 0$: the leading spatial-gradient stiffness vanishes. A higher-spatial-gradient EFT and its gravitational constraints must be supplied before that *exact* point is called perturbatively complete. The model's shift current obeys

$$\boxed{a^3 P_X \dot{\phi} = C_\phi, \qquad \phi = f\theta, \qquad a^3(2y-1)\sqrt{y} = (2y_0 - 1)\sqrt{y_0}, \quad a_0 = 1.} \tag{113}$$

Proposition G.2 (Early-radiation asymptotics of trial A [D]). *On the stable branch $y_0 > 1/2$ with nonzero conserved shift charge, the solution has $y(a) \sim [D/(2a^3)]^{2/3}$ as $a \to 0$, where $D =$*

$(2y_0 - 1)\sqrt{y_0}$. *Consequently* $\rho_\theta \propto a^{-4}$, $w \to 1/3$ *and* $c_s^2 \to 1/3$, *and the asymptotic* background *energy-density ratio relative to ordinary radiation is*

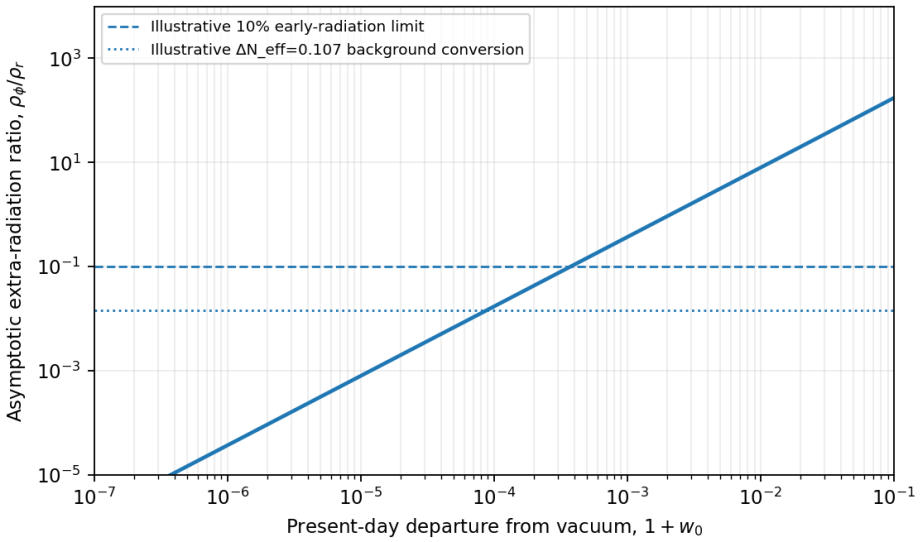
$$\boxed{\mathcal{R}_{\rm early} := \lim_{a\to 0} \frac{\rho_\theta}{\rho_r} = \frac{\Omega_{\theta 0}}{\Omega_{r0}} \frac{3}{y_0(3y_0-1)} \left[\frac{(2y_0-1)\sqrt{y_0}}{2}\right]^{4/3}}. \tag{114}$$

Proof. Equation (113) is monotonic for $y \geq 1/2$ and gives $2y^{3/2} \sim Da^{-3}$ at early times. With $\rho_\theta = M_*^4(-y + 3y^2) \sim 3M_*^4 y^2$, and $\rho_r = \rho_{r0} a^{-4}$, normalize $M_*^4 = \rho_{\theta 0}/[y_0(3y_0 - 1)]$ and simplify. □

For the *illustrative* present-day normalizations $(\Omega_{\theta 0}, \Omega_{r0}) = (0.70, 9 \times 10^{-5})$, the independently verified relation gives

y_0	w_0	$c_s^2(1)$	$\mathcal{R}_{\rm early}$
0.50001	−0.999920	0.000010	0.0127
0.50004	−0.999680	0.000040	0.0804
0.50010	−0.999200	0.000100	0.2727
0.50500	−0.961165	0.004926	48.64

Even the lenient *illustrative* requirement $\mathcal{R}_{\rm early} < 0.1$ forces $y_0 < 0.50004711$ and $1 + w_0 < 3.77 \times 10^{-4}$. A July 2026 BBN+CMB+BAO analysis reported an excess free-streaming radiation bound $\Delta N_{\rm eff} < 0.107$ (95% CL; `https://doi.org/10.1103/5vk8-v1rb`). Translated *only at the background energy-density level*, this would correspond to $\mathcal{R}_{\rm early} \approx 0.0144$ and $1 + w_0 \lesssim 8.8 \times 10^{-5}$. This is *not* a direct observational bound on the condensate, since its perturbations need not behave as freely streaming neutrinos; a dedicated Boltzmann likelihood analysis is required. The conservative 10% calculation alone demonstrates the strong early-radiation tension for appreciably evolving w_0.



Q058 numerical cross-check: early radiation ratio versus the present departure $1 + w_0$. The curves are model outputs, not measured data.

On the near-condensate stable branch $w + 1 \propto a^{-3}$, so a CPL-like local slope is $w_a \simeq 3(1 + w_0) > 0$. Thus the simplest shift-symmetric condensate tends toward $w = -1$ with the opposite sign from thawing ($w_a < 0$), while a substantial deviation today brings an excessive early-radiation background. It is not a successful explanation of evolving late-time dark energy in this minimal form.

G.5 Q058 trial B: a label-neutral hilltop with an explicit selector

A canonical scalar avoids the early-radiation behavior of the kinetic condensate. Let an additional positive internal metric H_0 be a *physical reference spurion*; define

$$\boxed{\mathcal{J}(\mathsf{H},\mathsf{H}_0) := \frac{1}{2}\,\mathrm{tr}(\mathsf{H}\mathsf{H}_0^{-1}) = \cosh[2(\theta-\theta_0)] \geq 1.} \tag{115}$$

The reference spurion and H may transform together under changes of internal labels, but fixing H_0 physically breaks the relative global boost shift. A simple, bounded, explicitly covariant-in-internal-labels potential is

$$\boxed{\Delta\Gamma_{\mathsf{H}}^{(B)} = \int \mathrm{d}^4x\sqrt{-g}\left[X_{\mathsf{H}} - \frac{U_0}{\mathcal{J}(\mathsf{H},\mathsf{H}_0)}\right], \quad U(\theta) = U_0\,\mathrm{sech}[2(\theta-\theta_0)].} \tag{116}$$

At the selected hilltop $\theta=\theta_0$ one has $U=U_0$, $U'=0$, $U''=-4U_0$, and physical scalar mass parameter $m_\theta^2 = -4U_0/f^2$. The homogeneous canonical scalar obeys $\rho_\theta + p_\theta = f^2\dot\theta^2 \geq 0$ and $c_s^2=1$; its tachyonic hilltop evolution takes place on long cosmological wavelengths if f is of order a few $M_{\rm Pl}$. A constant hilltop provides a mathematically exact $w=-1$ branch with $3M_{\rm Pl}^2H^2 = U_0$, but it is not a stable late-time minimum.

As one existence check, numerical integration from $a_i = 10^{-4}$ with $\dot\theta_i = 0$, $(\Omega_{m0},\Omega_{r0}) = (0.30, 9\times10^{-5})$, $f = 3M_{\rm Pl}$ and U_0 tuned so $H(1)=H_0$ yields

$\theta_i-\theta_0$	$U_0/(M_{\rm Pl}^2H_0^2)$	w_0	w_a (local)
0.08	2.1324	−0.99799	−0.00351
0.16	2.2308	−0.99243	−0.01295
0.24	2.3965	−0.98453	−0.02571
0.32	2.6320	−0.97568	−0.03908

The trial is a standard thawing canonical scalar in internally adapted coordinates; its potential, reference spurion, scales, and initial conditions are new hypotheses. It is not selected by QGD's two axioms, and the numbers are not a DESI/Planck/DES likelihood fit or a distinct QGD observable. It does, however, avoid picking a + versus − label *within its chosen block* and supplies a healthy short-wavelength conditional example; choosing the block itself is still additional information.

G.6 Q058 final selection verdict and consequences for canonical QGD

1. **Existing theory [D]/[C]:** The signed-sector geometry, metric-universal matter, quantum-state response, and boundary data do not select a metric-null dark-energy amplitude. The canonical renormalized Λ branch remains the defined late-time accelerating baseline.

2. **Extra sector law [A]:** A positive internal generalized metric with an action can create a physically observable modulus, but it is added structure and reduces local representation equivalence. Global-symmetry-preserving $P(X)$ has a stable-side branch and a $w=-1$ condensate limit; the exact limit needs higher-gradient control.

3. **New Q058 exclusion [D]/[C]:** With conserved shift charge, trial A becomes radiation-like at early times. An appreciable present deviation from $w=-1$ is incompatible with even a lenient early-radiation background budget in the specified cosmology. The elementary shift-symmetric condensate is not an adequate evolving-dark-energy solution.

4. **Conditional survivor [A]/[C]:** Trial B provides a bounded label-neutral internal hilltop and healthy canonical short-wavelength dynamics, at the cost of an explicit reference spurion and tuned potential scale. It is ordinary quintessence until an additional QGD-specific selection and observable consequence are established.

Canonical policy. Retain both founding axioms and the full master metric; leave the baseline action unchanged. Keep Q055–Q058 as researched, proof-tagged *extensions* in Chapter 8. Do not interpret a numerical $w(z)$ fit as a derivation of an additional QGD energy component, and do not count the same vacuum energy simultaneously in $\Lambda_{\rm ren}$, the Hartree determinant, and an added sector potential.

H Q061: action, symmetry and dark-energy status at the final review

Established homogeneous theory. On the regular, complete metric-only branch the classical Friedmann equations follow by variation of $\Gamma^{\rm ren}_{\rm met}[g(\Sigma)]$. The renormalized $\Lambda_{\rm ren}$ gives ordinary $w=-1$ dark energy but has an independently calibrated value. The conformal Dirac anomaly contributes a specified state-/scheme-dependent H^4 correction; its controlled GR-connected branch is negligible at late times. The algebraic self-accelerating $H^2\sim 1/\alpha$ branch lies outside the adopted low-energy EFT regime. Local R^2, C^2, E_4 contributions by themselves do not create a new controlled constant-curvature de Sitter branch when $\Lambda_{\rm ren}=0$ in four-dimensional regular metric-only QGD.

H.1 Only an explicitly physical sector can add dynamical dark energy

The signed-sector null pair can be rescaled by local $O(1,1)$ transformations without changing g. Hence neither the baseline action, quantum stress depending only on g, nor boundary data select that amplitude as a new energy component. A trial direct-sector action is a *new constitutive hypothesis* that reduces the representation freedom. The label-neutral Q058 construction uses a positive internal metric $\mathsf{H}(\theta)$ and a reference H_0 with

$$\mathcal{J}(\mathsf{H},\mathsf{H}_0)=\tfrac{1}{2}\,\mathrm{tr}(\mathsf{H}\mathsf{H}_0^{-1})=\cosh[2(\theta-\theta_0)]\geq 1,\qquad U(\theta)=\frac{U_0}{\cosh[2(\theta-\theta_0)]}. \tag{117}$$

The optional action is

$$\boxed{\Delta S_\theta=-\int d^4x\sqrt{-g(\Sigma)}\left[\frac{f^2}{2}(\nabla\theta)^2+U(\theta)\right].} \tag{118}$$

It is spacetime-scalar and preserves the common metric for ordinary matter if θ is physically defined as an internal scalar. Yet its new positive internal metric, reference H_0, kinetic scale f, and potential height U_0 are not selected by Axiom I or II; without a sector connection, its derivative term does not retain the unrestricted *local* $O(1,1)$ representation freedom. The fixed reference raises the same off-shell Noether consistency question as other external backgrounds. One must either specify a reduced physical sector theory with consistent constraints or give the new internal structures appropriate dynamical/transformational definitions. This is a consistent *conditional scalar-tensor extension*, not a term derived from the canonical metric-only action.

H.2 Cosmological phenomenology and energy accounting

The canonical scalar has $\rho_\theta=f^2\dot\theta^2/2+U$, $p_\theta=f^2\dot\theta^2/2-U$ and high-k $c_s^2=1$ for $f^2>0$. Its hilltop is long-wavelength unstable with mass parameter $m_\theta^2=-4U_0/f^2$; controlled thawing examples in Q055–Q058 give $w_0>-1$ and modest negative w_a, but no parameter-free prediction. The shift-symmetric condensate instead faces a strong additional early-radiation constraint and is not preferred as a robust explanation for appreciable late-time evolution. The θ stress is counted only once in the total Einstein/natural-invariant Euler equations, in addition to any independently retained $\Lambda_{\rm ren}$. The bounded hilltop is therefore preserved as an *optional research model*, not inserted into $\Gamma^{\rm ren}_{\rm QGD}$ as a discovered term.

Causal quantum cosmology. For time-dependent states the QN004 Euclidean form factors must be embedded in an in–in/closed-time-path renormalized action before inferring retarded cosmological response. A formally covariant local effective action is not automatically a complete real-time quantum prediction. These are the action-level lessons of Q059–Q060 and do not alter the previously calculated cosmological solutions in their stated approximation.

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